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Palaeomagnetism and $^{40}Ar/^{39}Ar$ geochronology of upper Palaeogene volcanic rocks from Central Tibet: implications for the Central Asia inclination anomaly, the palaeolatitude of Tibet and post-50 Ma shortening within Asia

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SUMMARY

Palaeomagnetic results from upper Palaeogene volcanics from three localities in the Qiangtang terrane of Central Tibet record a maximum likelihood palaeolatitude of 28.7 ± 3.7°N. These are the first volcanic-based palaeomagnetic results from the plateau interior that appear to average secular variation well enough to yield a reliable palaeolatitude estimate. This palaeolatitude estimate is significantly lower than the ∼36 ± 3°N predicted for Central Tibet by late Palaeogene Eurasia reference poles, consistent with the so-called Central Asia inclination anomaly, but it is consistent with values predicted from upper Palaeogene reference poles for Mongolia. The mean inclination of coeval sedimentary rocks from Central Tibet is ∼15 ± 5° shallower than our volcanic results, similar to the magnitude of sedimentary inclination shallowing estimated from Elongation/Inclination analysis of remanent directions in clastic rocks from the same region. We conclude that most of the inclination anomaly observed in volcanic rocks and that which remains after removal of sedimentary flattening in clastic rocks probably is due to the combined effect of reference poles that are not representative of stable Asia and by Cenozoic intracontinental shortening within Central Asia on the order of several hundred (not more than 1000) kilometers. Our compilation of palaeomagnetic data from the Qiangtang terrane further suggest that the southern margin of the Lhasa terrane (the southern margin of pre-India Asia) was located as far south as 20°N throughout Eocene time. These palaeolatitudes, in conjunction with the upper crustal shortening record for Asia, indicate that only approximately one third of the post-50 Ma India–Asia convergence is partitioned in Asian lithosphere. Low latitudes also imply that much of the proto-Tibetan plateau occupied southerly, tropical to subtropical latitudes, which has important implications for the weathering and erosion of the Tibetan plateau, palaeoaltimetry proxies and early Cenozoic climate dynamics.

Key words: Palaeomagnetism applied to tectonics; Palaeomagnetism applied to geological processes; Continental tectonics: compressional; Asia.

1 INTRODUCTION

North–south-directed convergence has dominated the Mesozoic and Cenozoic tectonics of Central Asia, making palaeomagnetism an obvious and informative tool for evaluating the latitudinal displacements of Asian upper crust. Western China consists of several crustal fragments, which, from south to north, include the Lhasa, Qiangtang, Qaidam, Tarim and Junggar terranes (Fig. 1, inset). More than four decades of palaeomagnetic research in China have provided a framework for estimating the Mesozoic and Cenozoic palaeogeography of the Eurasia landmass (McElhinny 1973; Dupont-Nivet *et al.* 2010b). Palaeomagnetic results from Cretaceous strata on these terranes (Enkin *et al.* 1992; Chen *et al.* 1993; Halim *et al.* 1998; Tan *et al.* 2010) record inclinations that are ∼10–12° more shallow than predicted from the reference apparent polar wander path (APWP) for Eurasia (Besse & Courtillot 2002; Torsvik *et al.* 2008). In terms of palaeolatitude, these shallow inclinations are interpreted to indicate that these crustal blocks were located farther south during the Cretaceous than expected from the Eurasia APWP and have moved northward throughout the Cenozoic as a result of the India–Asia collision, which is generally considered to have begun 50 ± 20 Ma (Patriat & Achache 1984; Rowley

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1996; Aitchison *et al*. 2007; Dupont-Nivet *et al.* 2010a; Najman *et al.* 2010). The amount of intra-Asia convergence implied by this interpretation is generally consistent with shortening estimates within thrust belts separating these terranes (Yin & Harrison 2000; van Hinsbergen *et al*. in review), although more robust structural and palaeomagnetic studies are certainly needed to evaluate these estimates with confidence.

One should be able to use palaeomagnetic studies of Cenozoic rocks in a similar way to gauge the distribution and magnitude of shortening both in space and time. The expectation is that observed palaeomagnetic inclinations become progressively less shallow with respect to the reference poles in younger rocks. Palaeomagnetic data from Cenozoic strata, however, consistently reveal inclinations that are as or more shallow than those from Cretaceous rocks when compared to the Eurasia reference poles (Cogné *et al.* 1999; Gilder *et al.* 2001; Liu *et al.* 2003; Yan *et al.* 2005). When the corresponding palaeolatitudes are considered, the amount of intra-Asia Cenozoic convergence and shortening implied by some of these studies is twice as much as that estimated since the Late Cretaceous. If accepted at face value, these results imply the following palaeogeographic reconstruction, for which there is no geological support: ∼1000 km of north–south extension between stable Eurasia and Central Asia from the Cretaceous to middle Palaeogene followed by more than 2000 km of convergence between Tarim and Siberia during the Eocene and earliest Neogene (Cogné *et al.* 1999). This discrepancy between inclinations predicted from the Eurasia APWP and those measured in Cenozoic rocks from Central Asia is referred to as the Central Asia inclination anomaly (Westphal 1993; Chauvin *et al.* 1996; Cogné *et al.* 1999; Hankard *et al.* 2007). Possible explanations for the inclination anomaly include tectonics (Cogné *et al.* 1999), inclination shallowing in sediments due to depositional and lithification processes (King 1955), or non-dipole geomagnetic fields (Si & Van der Voo 2001). Determining the source of the inclination anomaly has been hampered by the lack of a latitudinal distribution of robust palaeomagnetic data from Asian volcanics that characterize the time-averaged behaviour of the geomagnetic field (i.e. describe secular variation) and from coeval and collocated sediments.

Here we present geochronologic and palaeomagnetic results from three distinct upper Palaeogene (∼35 Ma) volcanic localities in Central Tibet. Until recently, palaeomagnetic studies of Palaeogene volcanic strata that demonstrably average secular variation have been completely lacking from the Tibetan orogen (*cf.* Hankard *et al*. 2007; Dupont-Nivet *et al.* 2010a,b; Lippert *et al*. 2010). Our results are the first data from upper Palaeogene volcanics from the Tibetan plateau, constitute the only volcanic-based palaeomagnetic data set from the Qiangtang terrane, and are one of the few studies of Palaeogene volcanics in Central Asia. These Qiangtang data are crucial for evaluating the palaeolatitude of Central Tibet, which can be directly compared to (i) palaeolatitude estimates calculated from coeval sedimentary palaeomagnetic data to assess the magnitude of sedimentary inclination shallowing and (ii) to palaeolatitude estimates for Southern and Northern Tibet to assess the first-order amount and distribution of upper-crustal shortening within the Tibetan orogen. We first describe results from our geochronologic, rock magnetic and palaeomagnetic studies. These results are then evaluated in the context of the Central Asia inclination anomaly. We conclude with a refined estimate for the amount of post-50 Ma upper crustal shortening within Central Asia and a revised palaeogeographic model for the Tibetan plateau.

2 GEOLOGICAL SETTING

The Qiangtang terrane is bounded to the south by the Jurassic–Cretaceous Bangong suture and to the north by the Triassic Jinsha suture (Fig. 1). The east–west trending 6000+ m high Tanggula Shan and the Mesozoic Qiangtang blueschist-bearing anticlinorium (Kapp *et al.* 2000) divide the terrane into a northern and southern region. Cenozoic sediments and volcanics are common in both regions, and often crop out in distinct belts (Turner *et al.* 1996; Horton *et al.* 2002; Chung *et al.* 2005; Ding *et al.* 2007; Wang *et al*. 2008). In this study, we examined volcanic rocks from three localities distributed across the Qiangtang terrane: Wulan Wula Hu (WWH) in the northern region and Ejumaima (EJU) and Nading Co (NDC) in the southern region (Fig. 1).

The WWH volcanics are ∼40 Ma in age and consist of sheeted high-potassium basalts that crop out as prominent cliff-formers and as numerous isolated buttes and volcanic necks that rise 100–300 m above the surrounding landscape (Wang *et al.* 2008). These lavas were erupted over an area of $>1000\,km^2$. Lavas are flat-lying and unconformably overlie strongly folded Jurassic and lower Palaeogene non-marine strata.

The volcanic rocks at EJU and NDC are described by Ding *et al.* (2007). EJU lavas consist of ∼34 Ma sheeted trachytes that crop out as isolated buttes that stand 20–100 m above the surrounding plain (Ding *et al.* 2007). Only the largest, northernmost outcrop is suitably exposed for palaeomagnetic sampling. Here, flow unit boundaries are distinct and indicate that the lavas are only gently tilted. Our two EJU sampling localities are within 0.5 km of each other, sample the same stratigraphic section of flows, and show only slight variation in flow attitude. Sampling sites EJU 1–9 have a mean strike and dip of 133°, 21°South ($n = 21$, α_{95} cone of confidence on flow poles: 4.8°), while sites EJU 10–14 strike 102° and dip 23°South ($n = 20$, $\alpha_{95} = 5.3°$). The mean lava attitude of all sites strikes 117° and dips 21°South ($n = 41$; $\alpha_{95} = 3.8°$). Although we did not observe EJU lavas in contact with other map units, Ding *et al*. (2007) infer that the lavas unconformably overlie steeply south dipping Eocene red beds of the Upper Muggar Unit of DeCelles *et al.* (2007a).

At NDC, sheets of ∼36 Ma basalt and trachyte crop out over a $>700\,km^2$ region and are well exposed in the footwalls of active west-dipping normal faults (Ding *et al.* 2007). The NDC lavas are regionally flat lying, but are tilted at our two sampling localities. The mean flow attitude for sites NDC 11–20 strikes 148° and dips 22° southwest ($n = 29$, $\alpha_{95} = 3.2°$) while at sites NDC 21–25, flows strike 002° and dip 15° east ($n = 2$). Although the tilt of sites NDC 21–25 may appear poorly constrained with only two attitude measurements, extensive outcrop benches here clearly indicate the lava flows are gently tilted toward the east. Furthermore, an eastward tilt will have little effect on the tilt-corrected inclination, and therefore also on the calculated palaeolatitude. The NDC lavas unconformably overlie strongly folded Triassic marine strata (Ding *et al.* 2007).

3 GEOCHRONOLOGIC METHODS

We conducted $^{40}Ar/^{39}Ar$ studies on two samples from the WWH sampling locality to confirm that the sampled outcrop is part of the regional ∼40 Ma Zhuerkenwula volcanic field (Wang *et al.* 2008) and not related to a younger volcanic episode in the region. Fine-grained whole-rock chips free of visible inclusions were wrapped in aluminum foil packets, stacked in an aluminum canister and

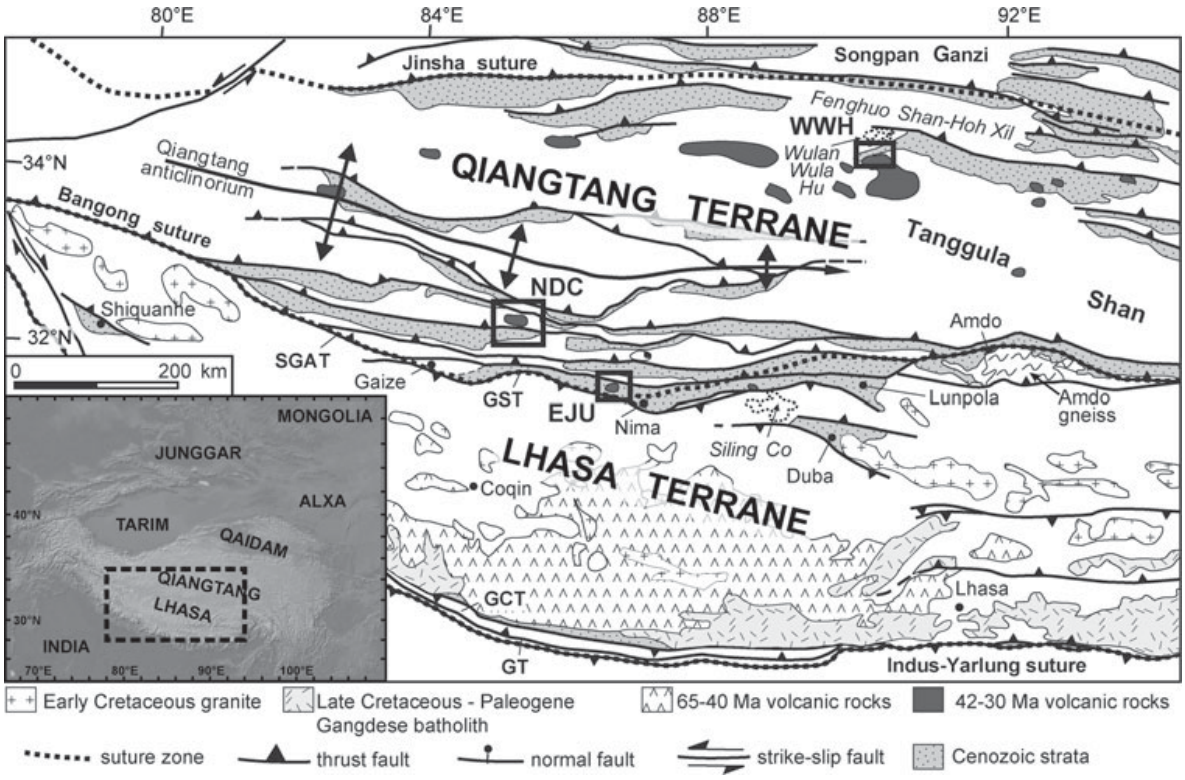


Figure 1. Map showing major tectonic features and distribution of Cretaceous and Cenozoic volcanism in Southern and Central Tibet. Sampling locations are indicated by black boxes: Wulan Wula Hu (WWH); Ejumaima (EJU); and Nading Co (NDC). Other abbreviations are: Gangdese thrust (GT); Great Counter thrust (GCT); Shiquanhe-Gaize-Amdo thrust system (SGAT) and the Gaize-Siling Co thrust (GST). Figure modified from Kapp *et al.* (2007a).

irradiated in the VT-C position of the Tsing-Hwa THOR research reactor for 30 hr. Three aliquots of the irradiation standard LP-6 biotite (Odin *et al.* 1982), which has a calibrated $^{40}Ar/^{39}Ar$ age of 129.3 ± 0.7 Ma based on Fish Canyon Sanidine (28.201 ± 0.046 Ma) (Baksi *et al.* 1996; Renne *et al.* 1998; Kuiper *et al*. 2008), were also irradiated. The neutron flux gradient across the top and bottom of the irradiation canisters was less than 0.9 per cent, as indicated by the variation of J-values from the flux monitors. After irradiation, the samples and standards were step heated using a double-vacuum Ta resistance furnace. The released gas was analyzed on a VG1200 mass spectrometer at the National Taiwan University. Measured isotopic ratios (Table A1 of Auxiliary Information) were corrected for the effects of mass discrimination, system blanks and interfering isotopes produced during irradiation using factors reported in Lo *et al*. (2002). Age calculations, discussed below, were made using a total decay constant of $(5.543 \pm 0.010) \times 10^{-10}$ a^{-1} (Steiger & Jager 1977). The quoted error for each apparent age is one standard deviation and includes the error in the J-value and the standard error, but not for the error in the interference corrections.

4 PALAEOMAGNETIC METHODS

Palaeomagnetic sampling consisted of drilling 6–8 2.5 cm diameter cores at each sampling site using a gasoline-powered drill. Sampling sites were selected based on field evidence for flow unit boundaries, using scoria layers, oxidation fronts and crystal size changes as guides. Samples were collected over the height of the flow unit and typically across 1–2 m of lateral outcrop. Each sample was oriented *in situ* using both magnetic and sun compasses. The average absolute difference between magnetic and sun compass readings is $2.5 \pm 4.8^{\circ}$ (2σ, $n = 161$) at all localities, so we feel confident including samples with no sun compass orientations in the results.

We conducted a series of rock magnetic experiments to determine the magnetic mineralogy and magnetic fabric of sample subsets from each locality. Magnetic hysteresis curves of small sample chips were measured on a Princeton Measurements MicroMag 2900 alternating gradient magnetometer to assess magnetic grain size. A Kappabridge KLY-2 susceptibility bridge with a CS-2 furnace that measures the temperature dependence of susceptibility was used to estimate the Curie temperatures of magnetic phases. Approximately 500 mg of bulk rock was crushed in a ceramic mortar and then heated in an argon environment. The anisotropy of magnetic susceptibility (AMS) of each sample was measured prior to demagnetization using a Kappabridge KLY-3 and analyzed using the PaleoMac software package (Cogné 2003).

All demagnetization treatments and remanence measurements were performed in a magnetically shielded room at the University of California, Santa Cruz. A subset of sample pairs from each locality was progressively demagnetized using thermal and alternating field (AF) treatments and the remanence after each step measured on a 2G cryogenic magnetometer controlled by software created in-house (Morris *et al.* 2009). Well-defined characteristic remanent magnetizations (ChRMs) isolated by the two treatment types are generally indistinguishable from each other (see Section 5.3), so all remaining samples were demagnetized using AF treatment. The majority of sample directions were determined using principal component analysis to obtain line-fits (Kirschvink 1980), and a few were best fit to remagnetization great circles (Halls 1978). All determined directions had mean angular deviations less than

15°, and most (>95 per cent) less than 10°, so no samples were excluded from further analysis because of poorly resolved directions. Site mean directions for sites that included directions fit by great circles were calculated following McFadden & McElhinny (1988). All other site and locality mean directions were calculated using Fisher statistics (Fisher 1953). Site mean directions were calculated using only AF-demagnetized ChRMs, and sample directions greater than two standard deviations away from the mean direction were excluded from the final mean. These calculations were made using the PalaeoMac software package (Cogné 2003). Following Johnson *et al*. (2008), site mean directions defined by fewer than five samples or by precision parameters (κ) less than 50 were excluded from further analysis. This filtering keeps site mean directions at comparable scatter and quality.

Several (e.g. <10) independent spot readings (i.e. distinct mean directions) are needed to accurately describe the time-averaged behaviour of the geomagnetic field. Because the length of time between the eruption of successive lava flows is often much less than the typical timescale of geomagnetic secular variation, each site at a specific sampling locality may not represent an independent sampling of the geomagnetic field. Indeed, visual inspection reveals that site mean directions at each sampling locality are clustered into a few groups (see below), consistent with short bursts of episodic eruptions. Therefore we have grouped site mean directions to reduce redundant sampling. Stratigraphically successive site mean directions are compared using a standard *F*-test for a common true mean direction (McFadden & Lowes 1981). Group mean directions for successive sites found to share a common direction were then calculated by taking the Fisher mean direction of all samples in the grouped sites.

Palaeomagnetic studies of the present and ancient magnetic field indicate that virtual geomagnetic poles (VGPs) fit a Fisherian distribution better than the parent directions (Creer 1962; Cox 1962; McElhinny & McFadden 1997; Kono 1997; Johnson *et al.* 2008). Therefore, we transform site mean and direction group directions into VGPs, calculate the Fisher mean VGP for each sampling locality from each data set, and then calculate the mean direction at each sampling locality following Butler (1992). Uncertainties on the mean inclination and declination follow Demarest (1983).

5 RESULTS

5.1 Argon-argon geochronology for WWH lavas

Apparent $^{40}Ar/^{39}Ar$ ages were calculated from both age spectra and inverse isochrons (Fig. 2 and Table A1). The age spectrum for Sample 06X036b, from the top of the sampled volcanic section at WWH displays a plateau defined over 86.7 per cent of the released argon, giving a plateau age of 38.4 ± 0.5 Ma (Fig. 2a). This spectrum also shows a typical recoil profile, however. For this reason, we calculated a total gas integrated age of 38.6 ± 0.5 Ma. For comparison, the inverse isochron age is 38.4 ± 0.5 Ma (Fig. 2b). The age spectrum for Sample 06X109a from the bottom of the sampled section displays a plateau defined over 69.3 per cent of the

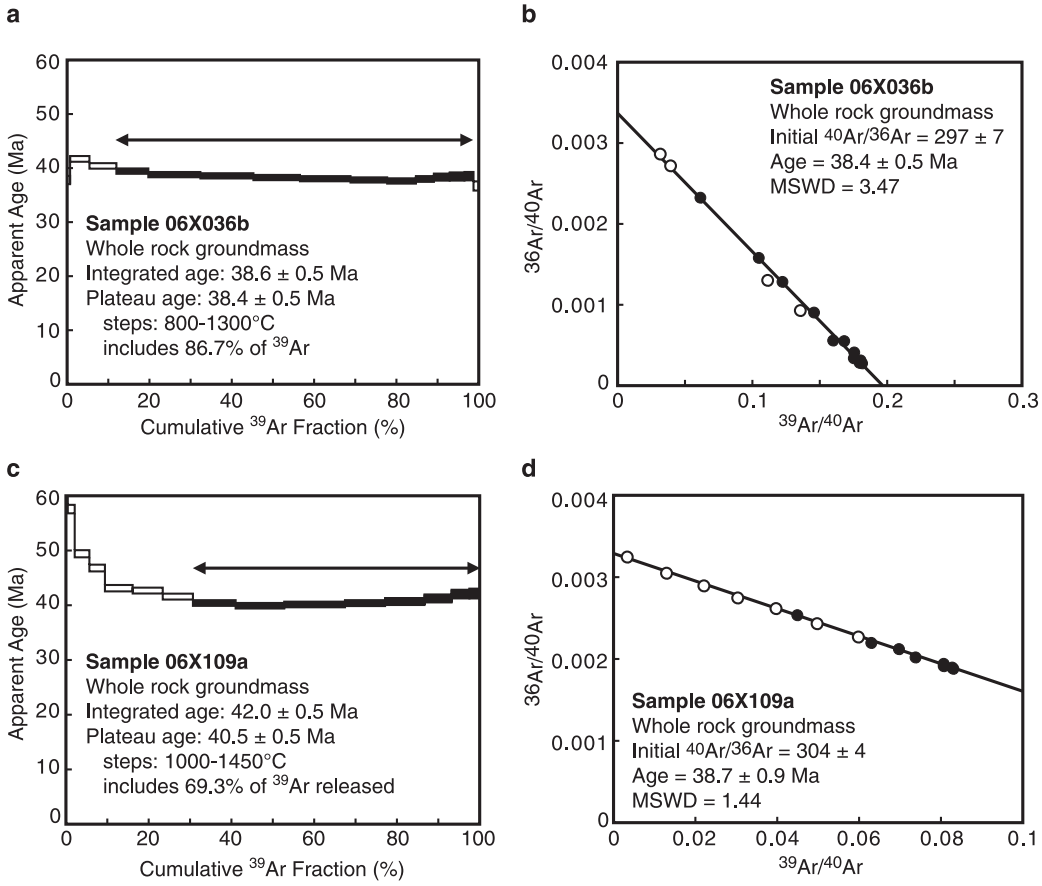


Figure 2. Argon age spectra and inverse isochron plots of Wulan Wula Hu basalts. Filled-in boxes and circles indicate the heating steps used to calculate the plateau age.

argon released, yielding a plateau age of 40.5 ± 0.5 Ma (Fig. 2c). Similar to the previous sample, the overall behaviour of the release spectra is consistent with significant argon recoil. We prefer the total-gas integrated age of 42.0 ± 0.5 Ma. The inverse isochron age for Sample 06X109a is 38.7 ± 0.9 Ma (Fig. 2d). The mean inverse isochron age of basalts at the WWH locality is 38.6 ± 0.5 Ma. These $^{40}Ar/^{39}Ar$ results yield similar ages to other basalts in the sampling region (Chung *et al.* 2005; Wang *et al.* 2008) and confirm that the WWH sampling locality is part of the east–west belt of upper Eocene potassic volcanic rocks north of the Tanggula Shan and not a younger belt of Hoh Xil lavas (Wang *et al.* 2008). Importantly, the WWH lavas are of similar age to the lavas sampled at EJU and NDC.

5.2 Rock magnetism

Eocene lavas at WWH typically display upright, square-to-slightly pot-bellied magnetic hysteresis loops. Squareness (M_{rs}/M_s) and coercivity ratios (H_{cr}/H_c) indicate a pseudo-single domain (PSD) mean magnetic grain size (Fig. 3a), but the ratios for the sample collection plot along a mixing curve between multidomain (MD) and single domain (SD) magnetite (Dunlop 2002). These results cannot distinguish between a distribution of grains clustered around large PSD sizes or a mixture of grains with mainly MD and some SD sizes. Magnetic hysteresis loops from EJU and NDC samples are similar to those from WWH, but with greater single-domain components if they do represent MD–SD mixtures (Figs 3b and c).

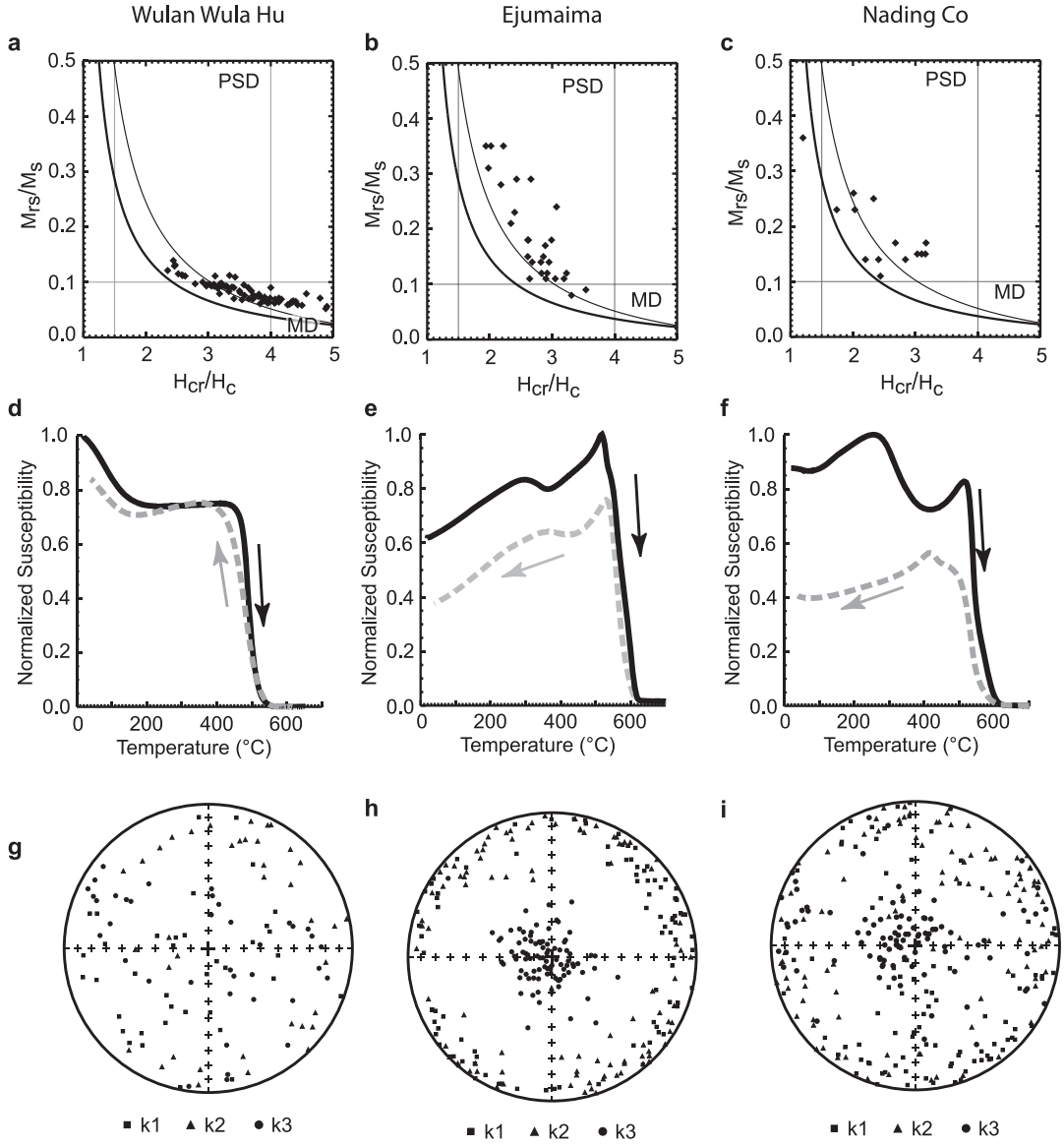


Figure 3. Rock magnetic properties of Wulan Wula Hu (WWH), Ejumaima (EJU) and Nading Co (NDC) lavas. (a)–(c) Day plot of hysteresis data, showing the magnetic grain size of bulk samples. Thin straight grey lines delineate magnetic grain size fields: pseudo-single domain (PSD) and multidomain (MD). Thick curved solid line is a theoretical mixing curve of MD and uniaxial single-domain (SD) magnetite; thin curved line is a mixing curve of MD and cubic SD magnetite, from Dunlop (2002). (d)–(f) Normalized susceptibility *v.* temperature results; all samples heated in an argon environment. (d): Sample 06X061 (WWH Site 4). (e) Sample 07EJ005 (EJU Site 1). (f) Sample 07ND151 (NDC Site 17). (g)–(i) Equal area net showing the distribution of the normalized eigen values of the anisotropy of magnetic susceptibility (AMS) tensor: maximum (k_1, squares), median (k_2, triangles) and minimum (k_3, circles) eigen-values projected onto the lower hemisphere of the net.

Susceptibility versus temperature (κ-T) experiments for WWH basalts are characterized by reversible heating and cooling curves (Fig. 3d), whereas EJU and NDC lavas lose ∼50 per cent of their initial room-temperature susceptibility after a heating and cooling cycle (Figs 3e and f). All three localities have dominant Curie points between 500 and 580 °C, consistent with iron-rich titanomagnetite. They also exhibit lower Curie points in the 175–350 °C range, suggesting a minor coexisting phase with higher titanium content.

Although an obvious flow-parallel magnetic fabric characterizes the lavas at EJU (Fig. 3h) and NDC (Fig. 3i), the mean lineation (L) and foliation (F) parameters for each locality are low and are not significantly different from each other (Table A2). Moreover, the maximum observed anisotropy (per cent h, Tauxe *et al.* 1990) is everywhere <3.1 per cent (Table A2). We also observe that the mean k_3 direction is not correlated with the mean remanence direction. Thus, the magnetic fabric observed in these lavas is unlikely to bias the ChRM directions to a degree that is important to this tectonic study.

5.3 Demagnetizations and site mean directions

5.3.1 Wulan Wula Hu

Figs 4(a)–(d) show typical demagnetization behaviour of WWH lavas. AF and thermal demagnetization treatment of some samples removes a low coercivity, low temperature component that is north and down. It is similar to the modern field direction in this region, which is about inclination 50° and declination 0° (Fig. 4e), and is probably a weak viscous remanent magnetization. Samples from the stratigraphically highest sites (WWH Sites 1, 2 from near the top of a northeast-facing bluff) have the highest NRM intensities as well as low-coercivity directions that are east and up (Fig. 4a). They also have a lower median destructive field (MDF) of 8 ± 2 mT than samples from stratigraphically lower sites (18 ± 5 mT), suggesting that these directions may record a lightning-induced isothermal remanent magnetization. About one third of the samples have low-coercivity directions with declinations ranging from northwesterly to southwesterly and inclinations that are generally shallower than the modern field inclination. These may also be due to lightning or weak viscous overprints of other stray fields and are easily removed by AF treatment.

A consistent ChRM that is north and down was indicated in 78 of 84 samples (Fig. 4f), with almost all samples displaying linear trends toward the origin during further demagnetization once the overprints were removed (Figs 4b–d). WWH Sites 1 and 2 display three components (Fig. 4a) and several of these sample directions are best fit using remagnetization circles (Halls 1978). Samples from the remaining sites have ChRMs defined by tight-fitting lines with five or more consecutive treatment steps between 20 and 180 mT (e.g. Figs 4a,b and d; Table A3a).

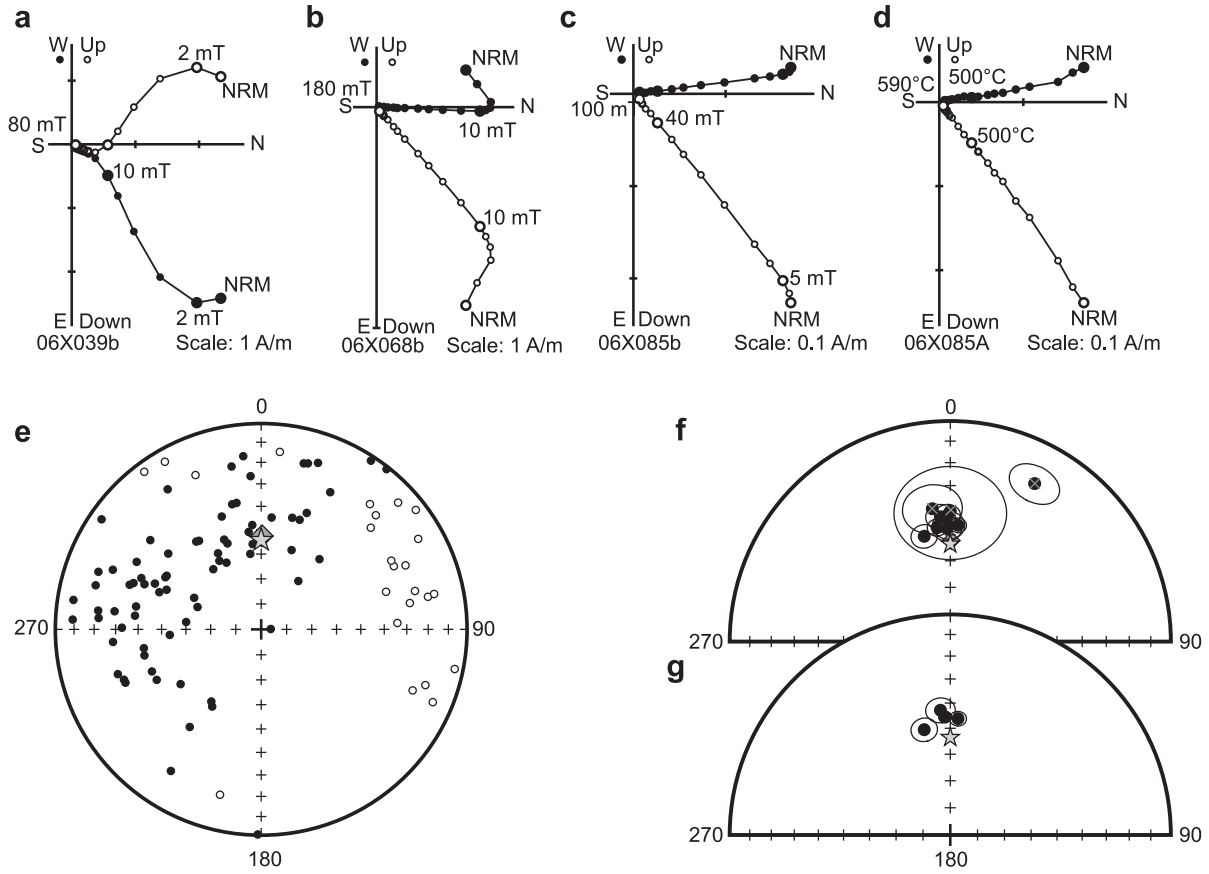


Figure 4. Palaeomagnetic results from Wulan Wula Hu lavas. Representative alternating field (AF) (a)–(c) and thermal (d) demagnetization results in *in situ* coordinates. Sample number is indicated below each orthogonal vector plot. (e) Equal area net of low-coercivity overprints. Closed (open) symbols are projected on the lower (upper) hemisphere. The International Geomagnetic Reference Field (IGRF) direction and dipole field direction are shown by the medium-grey diamond and light-grey star, respectively. (f) Site mean directions with 95 per cent confidence intervals. Site means rejected because of $n < 5$ or $\kappa < 50$ are indicated by grey crosses. (g) Group directions with 95 per cent confidence intervals.

Table 1 and Fig. 4(f) show the site mean directions for WWH lavas. Sites 1, 2 and 3 were discarded due to the low precision (κ) of their site mean directions. Notably, these are the topographically highest sampling sites and are characterized by directions best fit by great circles. We interpret the low precision of these site mean directions to reflect a persistent, unremoved overprint in individual samples, probably caused by lightening-induced saturation isothermal remagnetizations. Of the six remaining sites, we find only four independent direction groups, yielding a locality mean VGP at 82.1°N, 298.4°E ($A_{95} = 7.8°$, $K = 141.2$, $S = 6.8°$). The corresponding site mean direction at WWH is 46.1 ± 7.6° inclination, 355.8 ± 7.0° declination.

The WWH lavas are flat lying, so no tilt correction is applied. Given the hysteresis and κ-T results that indicate remanence is carried mainly by unaltered, PSD titanomagnetite with substantial MDFs around 18 mT, the univectorial decay to the origin during demagnetization of most samples after the removal of overprints, and the fact that both AF and thermal treatments of paired samples isolate the same ChRM directions, we conclude that these directions are carried by primary thermoremanence that has reliably recorded the ancient field.

5.3.2 Ejumaima

AF demagnetization of EJU volcanics revealed a stable remanence with MDF of 44 ± 14 mT and, typically, two directional components (Figs 5a–d and Table A3b). The first is a low-coercivity (<10 mT) direction that tends to cluster around the modern field direction in geographic coordinates and is probably a modern-field viscous overprint (Fig. 5e). It is distinct and cleanly separated from the high-coercivity ChRM component (50–180 mT) that is southeast and up (Fig. 5f). Thermal demagnetization of selected sister-samples isolates the same ChRM direction as AF demagnetization (Figs 5b and c). Ninety-six per cent of the samples studied yielded interpretable ChRMs. The mean direction for Site 12 was excluded from further analysis because it is defined by only four samples. Site mean directions are tightly clustered and can be binned into seven independent direction groups (Figs 5f–i, Table 1), yielding an *in situ* mean VGP located at 60.0°N, 6.3°E ($A_{95} = 6.3°$, $K = 93.4$, $S = 8.4°$). The corresponding *in situ* locality mean direction at EJU is −51.4 ± 5.5° inclination, 144.4 ± 6.0° declination.

The sublocality-specific flow attitudes at EJU are very similar and the overall tilt of the lavas is gently toward the south, consistent with the location of EJU lavas in the proximal footwall of the south-directed Muggar thrust (Kapp *et al*. 2007a). We apply a tilt correction that utilizes the regional tilt of the lavas at all 14 sampling sites to reflect slight differences in the initial slope onto which the lavas flowed and the regional tilting of the entire volcanic unit following eruption. This tilt correction results in a Fisher mean VGP of direction groups at 30.0°N, 21.1°E ($A_{95} = 6.6°$, $K = 85.0$, $S = 8.8°$). The corresponding tilt-corrected locality mean direction at EJU is −54.1 ± 5.4° inclination, 106.5 ± 6.4° declination.

The straightforward rock magnetic characteristics and demagnetization behaviour of the EJU lavas indicate that the ChRM directions are primary thermoremanent directions. The 34.2 ± 1.3 Ma $^{40}Ar/^{39}Ar$ potassium feldspar age (Ding *et al*. 2007) and reversed polarity ChRM directions observed in these lavas are consistent with eruption and magnetization during the long reverse Chron C13r (Luterbacher *et al.* 2004). Thus, we have no reason to interpret the EJU ChRMs as secondary overprints.

5.3.3 Nading Co

Like lavas from WWH and EJU, NDC samples typically show two directional components with both thermal and AF demagnetization treatments (Figs 6a–d). The mean MDF for NDC samples is 35 ± 18 mT. Approximately one half of the samples collected has a north and down component that is removed by AF treatment up to 10 mT. Because these directions cluster near the modern field direction (Fig. 6e), we interpret this component to be a weak viscous overprint by the modern geomagnetic field.

Ninety-five per cent of the NDC samples yielded high-coercivity ChRM directions that are clearly indicated on vector demagnetization plots (Figs 6a–d and Table A3c). Thermal demagnetization gave the same results as AF on paired samples (Figs 6a and b). The mean direction for Site 12 was excluded from further analysis based on our filtering criteria described in Section 4. The thirteen remaining sites can be binned into nine independent mean directions (Table 1; Figs 6f–i), resulting in a Fisher mean VGP at 63.0°N, 213.3°E ($A_{95} = 6.1°$, $K = 73.0$, $S = 9.5°$) in *in situ* coordinates. The corresponding *in situ* mean direction at NDC is 27.1 ± 8.2° inclination, 21.8 ± 5.0° declination.

The tilts of 15° and 22° are too steep to be solely due to initial tilt of the surface onto which the low-viscosity basaltic lavas flowed, especially given that lavas are flat-lying in the eastern part of this volcanic field. Instead, these tilts are most likely related to the local active normal faults adjacent to the sampling localities (Ding *et al.* 2007). The Fisher mean VGP of the tilt-corrected direction groups is at 73.1°N, 207.6°E ($A_{95} = 8.8°$, $K = 35.4$, $S = 13.7°$, Table 1), resulting in a tilt-corrected mean direction at NDC of 40.0 ± 9.7° inclination, 15.5 ± 9.7° declination. No statistically robust or meaningful fold test is possible given so few independent directions from each tilted block and that the difference in structural attitudes is comparable with directional differences expected from secular variation. The straightforward rock magnetic characteristics and demagnetization behaviour, however, make us confident that the ChRM of these samples carry reliable records of the ancient field direction during primary cooling.

5.4 Reliability of the characteristic directions

In summary, rock magnetic experiments and demagnetization treatments indicate ChRM directions are carried by PSD low-Ti titanomagnetite and are not significantly biased by magnetic fabrics. Low demagnetization treatment (<15 mT; <300 °C) distinguishes a variety of viscous overprints from a consistent ChRM direction that is typically defined by line fits over at least five AF or thermal treatment levels. Furthermore, all samples have been fully demagnetized using at least 12 treatment levels. Thermally and AF-demagnetized ChRM directions from sample pairs are indistinguishable from each other, with a mean absolute difference between thermal and AF-derived inclinations of 2.2 ± 3.4° (2σ). The results from one treatment type are not systematically different from those of the other. These observations support the conclusion that the ChRM directions record a primary thermoremanence and are not biased by unremoved viscous overprints.

5.5 Qiangtang volcanic inclination data

The 38 sampling sites from the three localities investigated in this study have been binned into twenty stratigraphically distinct direction groups. As explained above, the reason for binning site mean

Table 1. Site and group mean directions for Qiangtang Lavas.

Direction group	Site	Coordinates		Flow attitude		n	N	LF	GC	NF	D_g	I_g	D_s	I_s	κ	α_{95}
		Lat. (°N)	Long. (°E)	Strike	Dip											
Wulan Wula Hu [38.6 ± 0.5 Ma mean inverse isochron $^{40}Ar/^{39}Ar$ age]																
–	~~Site 1~~	~~34.536~~	~~90.227~~	~~0~~	~~0~~	~~10~~	~~11~~	~~4~~	~~5~~	~~2~~	~~28.1~~	~~19.9~~	~~28.1~~	~~19.9~~	~~39.1~~	~~8.1~~
–	~~Site 2~~	~~34.536~~	~~90.227~~	~~0~~	~~0~~	~~7~~	~~7~~	~~6~~	~~1~~	~~0~~	~~360.0~~	~~40.6~~	~~360.0~~	~~40.6~~	~~11.3~~	~~18.9~~
–	~~Site 3~~	~~34.536~~	~~90.227~~	~~0~~	~~0~~	~~9~~	~~10~~	~~8~~	~~2~~	~~0~~	~~352.3~~	~~39.1~~	~~352.3~~	~~39.1~~	~~28.2~~	~~10.0~~
W1	Site 4	34.536	90.227	0	0	11	11	11	0	0	355.5	42.7	355.5	42.7	83.6	5.0
W2	Site 5	34.536	90.227	0	0	7	7	7	0	0	346.1	49.3	346.1	49.3	170.6	4.6
W3	Site 6	34.536	90.227	0	0	6	6	6	0	0	3.8	46.1	3.8	46.1	543.5	2.9
W4	Site 7	34.536	90.227	0	0	7	7	7	0	0	353.4	46.8	353.4	46.8	322.0	3.4
W4	Site 8	34.536	90.227	0	0	7	7	7	0	0	359.2	46.2	359.2	46.2	120.5	5.5
W4	Site 9	34.536	90.227	0	0	7	10	6	2	2	358.4	47.4	358.4	47.4	152.8	5.0
W4	Site 10	34.536	90.227	0	0	8	8	8	0	0	358.8	42.1	358.8	42.1	164.4	4.3
Grouped directions																
W1	Site 4	34.536	90.227	0	0	11	11	11	0	0	355.5	42.7	355.5	42.7	83.6	5.0
W2	Site 5	34.536	90.227	0	0	7	7	7	0	0	346.1	49.3	346.1	49.3	170.6	4.6
W3	Site 6	34.536	90.227	0	0	6	6	6	0	0	3.8	46.1	3.8	46.1	543.5	2.9
W4	Sites 7–10	34.536	90.227	0	0	29	32	28	1	0	357.5	45.5	357.5	45.5	154.5	2.2
Locality mean VGP of direction groups and observed direction						**n**	**P-lat. (°N)**	**P-long. (°N)**	**K**	**A95**	**S**	**I ± ΔI**	**D ± ΔD**			
	WWH	34.5	90.2			4	82.1	298.4	141.2	7.8	6.8	46.1 ± 7.6	355.8 ± 7.0			
Ejumaima [34.2 ± 1.3 Ma mean inverse isochron $^{40}Ar/^{39}Ar$ age (Ding *et al*. 2007)]																
Locality E-A																
E1	Site 1	32.219	86.630	117	21*S*	8	8	8	0	0	135.9	−42.0	115.8	−45.3	173.2	4.2
E2	Site 2	32.219	86.630	117	21*S*	8	8	8	0	0	139.9	−47.9	114.9	−51.8	124.1	5.0
E2	Site 3	32.219	86.630	117	21*S*	9	9	9	0	0	138.9	−47.7	114.2	−51.3	88.9	5.5
E3	Site 4	32.219	86.630	117	21*S*	6	6	6	0	0	135.2	−55.7	103.2	−56.5	188.3	4.9
E4	Site 5	32.219	86.630	117	21*S*	7	7	7	0	0	153.1	−50.2	125.4	−58.5	165.4	3.7
E4	Site 6	32.219	86.630	117	21*S*	7	7	7	0	0	161.4	−49.5	135.1	−60.8	111.1	5.8
E4	Site 7	32.219	86.630	117	21*S*	7	7	7	0	0	156.1	−46.6	132.1	−56.5	405.4	3.0
E4	Site 8	32.219	86.630	117	21*S*	6	7	5	1	1	152.6	−46.7	128.3	−55.3	120.9	6.2
E4	Site 9	32.219	86.630	117	21*S*	6	6	6	0	0	149.9	−46.0	126.2	−53.8	128.2	5.9
Locality E-B																
E5	Site 10	32.219	86.805	117	21*S*	6	7	6	0	1	158.5	−52.9	127.9	−62.6	52.6	9.3
E5	Site 11	32.219	86.805	117	21*S*	7	7	7	0	0	151.7	−55.5	117.7	−62.2	219.2	4.1
–	~~Site 12~~	~~32.219~~	~~86.805~~	~~117~~	~~21 S~~	~~4~~	~~7~~	~~5~~	~~0~~	~~2~~	~~156.4~~	~~−57.2~~	~~119.7~~	~~−65.2~~	~~1450.0~~	~~2.4~~
E6	Site 13	32.219	86.805	117	21*S*	7	7	7	0	0	142.1	−56.0	108.5	−59.2	229.4	4.0
E7	Site 14	32.219	86.805	117	21*S*	7	7	7	0	0	152.0	−53.4	120.7	−60.7	434.7	2.9
Grouped directions																
E1	Site 1	32.219	86.630	117	21*S*	8	8	8	0	0	135.9	−42.0	115.8	−45.3	173.2	4.2
E2	Sites 2–3	32.219	86.630	117	21*S*	17	17	17	0	0	139.4	−47.8	114.5	−51.6	109.3	3.4
E3	Site 4	32.219	86.630	117	21*S*	6	6	6	0	0	135.2	−55.7	103.2	−56.5	188.3	4.9
E4	Sites 5–9	32.219	86.630	117	21*S*	33	34	32	1	1	154.8	−48.0	129.5	−57.2	145.5	2.1
E5	Sites 10–11	32.219	86.805	117	21*S*	12	14	13	0	1	151.6	−54.2	119.4	−61.2	266.9	2.7
E6	Site 13	32.219	86.805	117	21*S*	7	7	7	0	0	142.1	−56.0	108.5	−59.2	229.4	4.0
E7	Site 14	32.219	86.805	117	21*S*	7	7	7	0	0	152.0	−53.4	120.7	−60.7	434.7	2.9

Table 1. (*Continued.*)

Direction group	Site	Coordinates Lat. (°N)	Coordinates Long. (°E)	Flow attitude Strike	Flow attitude Dip	*n*	*N*	LF	GC	NF	D_g	I_g	D_s	I_s	κ	α_{95}
Locality mean VGP of direction groups and observed direction						***n***	**P-lat. (°N)**	**P-long. (°N)**	**K**	**A95**	**S**	**I ± ΔI**	**D ± ΔD**			
	EJU	32.2	86.8	*in situ coordinates*		7	60.0	6.3	93.4	6.3	8.4	−51.4 ± 5.5	144.4 ± 6.0			
	EJU	32.2	86.8	*tilt-corrected coordinates*		7	30.0	21.1	85.0	6.6	8.8	−54.1 ± 5.4	106.5 ± 6.4			
Nading Co [35.4 ± 2.4 Ma mean inverse isochron $^{40}Ar/^{39}Ar$ age (Ding *et al.* 2007)]																
Locality N–A																
N1	Site 11	32.737	85.575	148	22 *SW*	6	6	6	0	0	17.4	29.2	5.3	44.5	108.1	6.5
~~–~~	~~Site 12~~	~~32.737~~	~~85.575~~	~~148~~	~~22 SW~~	~~7~~	~~7~~	~~7~~	~~0~~	~~0~~	~~20.1~~	~~29.1~~	~~8.4~~	~~45.1~~	~~46.1~~	~~9.0~~
N2	Site 13	32.737	85.575	148	22 *SW*	6	7	7	0	0	28.1	38.1	13.7	55.9	124.3	6.0
N3	Site 14	32.737	85.575	148	22 *SW*	5	8	8	0	0	30.7	27.5	21.9	46.3	168.0	5.9
N4	Site 15	32.737	85.575	148	22 *SW*	8	8	8	0	0	30.2	15.3	24.9	34.4	72.5	6.5
N4	Site 16	32.737	85.575	148	22 *SW*	7	7	7	0	0	25.2	20.7	17.6	38.5	157.3	4.8
N5	Site 17	32.737	85.575	148	22 *SW*	7	8	8	0	0	24.9	27.7	14.8	45.2	247.0	3.8
N6	Site 18	32.737	85.575	148	22 *SW*	6	7	6	0	1	3.2	31.2	348.7	41.7	140.2	5.7
N7	Site 20	32.737	85.575	148	22 *SW*	7	7	7	0	0	27.4	13.8	22.2	32.2	208.9	4.2
Locality N–B																
N8	Site 21	32.685	85.604	002	15 *E*	8	8	8	0	0	17.4	33.2	25.8	28.1	128.2	4.9
N8	Site 22	32.685	85.604	002	15 *E*	5	8	6	1	1	17.0	37.6	26.9	32.4	104.3	7.8
N8	Site 23	32.685	85.604	002	15 *E*	5	6	3	2	1	24.5	34.3	32.6	27.5	90.4	8.7
N8	Site 24	32.685	85.604	002	15 *E*	6	6	6	0	0	21.0	28.3	28.2	22.5	502.1	3.0
N9	Site 25	32.685	85.604	002	15 *E*	6	6	6	0	0	15.9	21.5	20.7	17.2	269.9	4.1
Grouped directions																
N1	Site 11	32.737	85.575	148	22 *SW*	6	6	6	0	0	17.4	29.2	5.3	44.5	108.1	6.5
N2	Site 13	32.737	85.575	148	22 *SW*	6	7	7	0	0	28.1	38.1	13.7	55.9	124.3	6.0
N3	Site 14	32.737	85.575	148	22 *SW*	5	8	8	0	0	30.7	27.5	21.9	46.3	168.0	5.9
N4	Sites 15–16	32.737	85.575	148	22 *SW*	15	15	15	0	0	27.9	17.8	21.6	36.3	85.8	4.2
N5	Site 17	32.737	85.575	148	22 *SW*	7	8	8	0	0	24.9	27.7	14.8	45.2	247.0	3.8
N6	Site 18	32.737	85.575	148	22 *SW*	6	7	6	0	1	3.2	31.2	348.7	41.7	140.2	5.7
N7	Site 20	32.737	85.575	148	22 *SW*	7	7	7	0	0	27.4	13.8	22.2	32.2	208.9	4.2
N8	Sites 21–24	32.685	85.604	002	15*E*	24	28	23	3	2	19.7	32.9	27.9	27.2	118.9	2.7
N9	Site 25	32.685	85.604	002	15*E*	6	6	6	0	0	15.9	21.5	20.7	17.2	269.9	4.1
Locality mean VGP of direction groups and observed direction						***n***	**P-lat. (°N)**	**P-long. (°N)**	**K**	**A95**	**S**	**I ± ΔI**	**D ± ΔD**			
	EJU	32.7	85.6	*in situ coordinates*		9	63.0	213.3	73.0	6.1	9.5	27.1 ± 8.2	21.8 ± 5.0			
	EJU	32.7	85.6	*tilt-corrected coordinates*		9	73.1	207.6	35.4	8.8	13.7	40.0 ± 9.7	15.5 ± 7.6			

Notes: Statistics were calculated using the PalaeoMac software package (Cogné, 2003). *n*: number of samples used to calculate mean directions; *N*: numbers of samples demagnetized; LF, GC, NF: number of directions determined using line fits, great circle fits, and no fit possible, respectively; D_g, I_g (D_s, I_s): directions in geographic (stratigraphic) coordinates; κ: best estimate of Fisher's (1953) precision parameter for the mean direction; a_{95}: half angle of the 95 per cent confidence cone on the mean direction. See text for details on direction groups.

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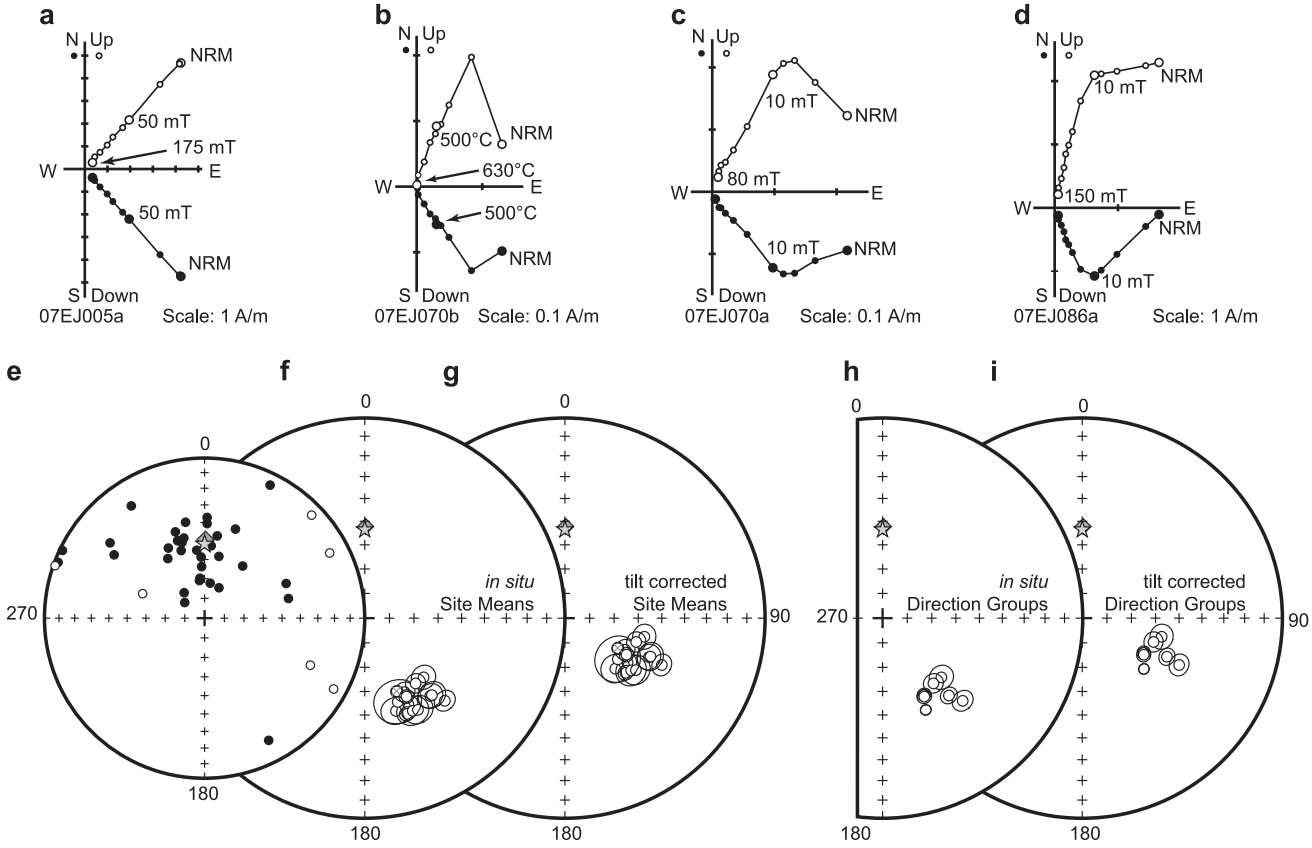


Figure 5. Palaeomagnetic results from Ejumaima lavas. Representative AF (a), (c)–(d) and thermal (b) demagnetization results in *in situ* coordinates. (e) Equal area net of low-coercivity overprints. (f) Equal area net showing site mean directions in *in situ*, and tilt-corrected coordinates. (g) Group directions with 95 per cent confidence intervals shown in *in situ* (h) and tilt-corrected (i) coordinates. Conventions and symbols are the same as in Fig. 4.

directions into direction groups is because stratigraphically successive flows with very similar directions were probably erupted close together in time and are not independent samplings of secular variation. If such flow units are not binned, then the number of directions used to calculate the Fisher mean VGP is spuriously high, and therefore the confidence limit on the mean will be spuriously low. The grouped and ungrouped mean directions may or may not be significantly different from each other, depending on how evenly or unevenly the independent directions are oversampled. In what follows, we calculate the palaeolatitude suggested by the recorded inclinations assuming a geocentric axial dipole (GAD). We make these calculations using both site mean directions and direction groups to assess the effect of binning.

We use inclination-only (and corresponding palaeolatitude-only) statistics to evaluate our palaeomagnetic data set because magnetic declinations vary considerably between localities. Variable declination values between localities, especially the apparent counterclockwise rotation of EJU with respect to the other sampling locations, is consistent with vertical-axis rotation of the Nima basin due to sinistral slip on basin-bounding faults and north–south-directed thrusting in the Nima region (Taylor *et al.* 2003; Kapp *et al.* 2007a). The use of inclination-only statistics does not compromise the utility of palaeomagnetic data from the south Qiangtang lavas in palaeogeographic reconstructions considered here because this study concerns only latitudinal displacements.

To calculate the mean palaeolatitude of the south Qiangtang lavas, we transform each site mean and direction group directions to its corresponding VGP. The inclination and corresponding uncertainty for a common site in southern Qiangtang (33°N, 88°E) is then calculated for each VGP. The corresponding palaeolatitude is calculated assuming a GAD field

$$\lambda_v = \tan^{-1}[\tan(I_v)/2], \quad (1)$$

where λ_v is the virtual palaeolatitude of the sampling site and I_v is the observed site mean or direction group inclination at the reference site from each VGP. The maximum likelihood palaeolatitude (λ_{MLE}) of these virtual palaeolatitudes is calculated following the procedure of Arason & Levi (2010). Uncertainties in inclinations and palaeolatitudes are calculated following Demarest (1983) and Arason & Levi (2010). The results are shown in Table 2 and do not differ significantly from values calculated with the earlier maximum likelihood method of McFadden & Reid (1982).

The palaeolatitude calculated from the tilt-corrected site mean directions is 28.9 ± 3.0°N, while from the direction groups it is 28.7 ± 3.7°N. Thus, for our palaeomagnetic data set, direction groups return mean palaeolatitudes that are essentially the same as the palaeolatitude calculated from ungrouped sites. It may appear from this analysis that binning palaeomagnetic data has no significant effect on the latitude calculations, but we emphasize that this procedure more accurately characterizes the geomagnetic field behaviour. Notably, in our Qiangtang example, the uncertainty on the maximum likelihood palaeolatitude is slightly larger for the direction groups than site means. Given a different volcanic-based palaeomagnetic data set, the effects of binning may (or may not) be more evident (e.g. Jarboe *et al*. 2008; Chenet *et al*. 2009).

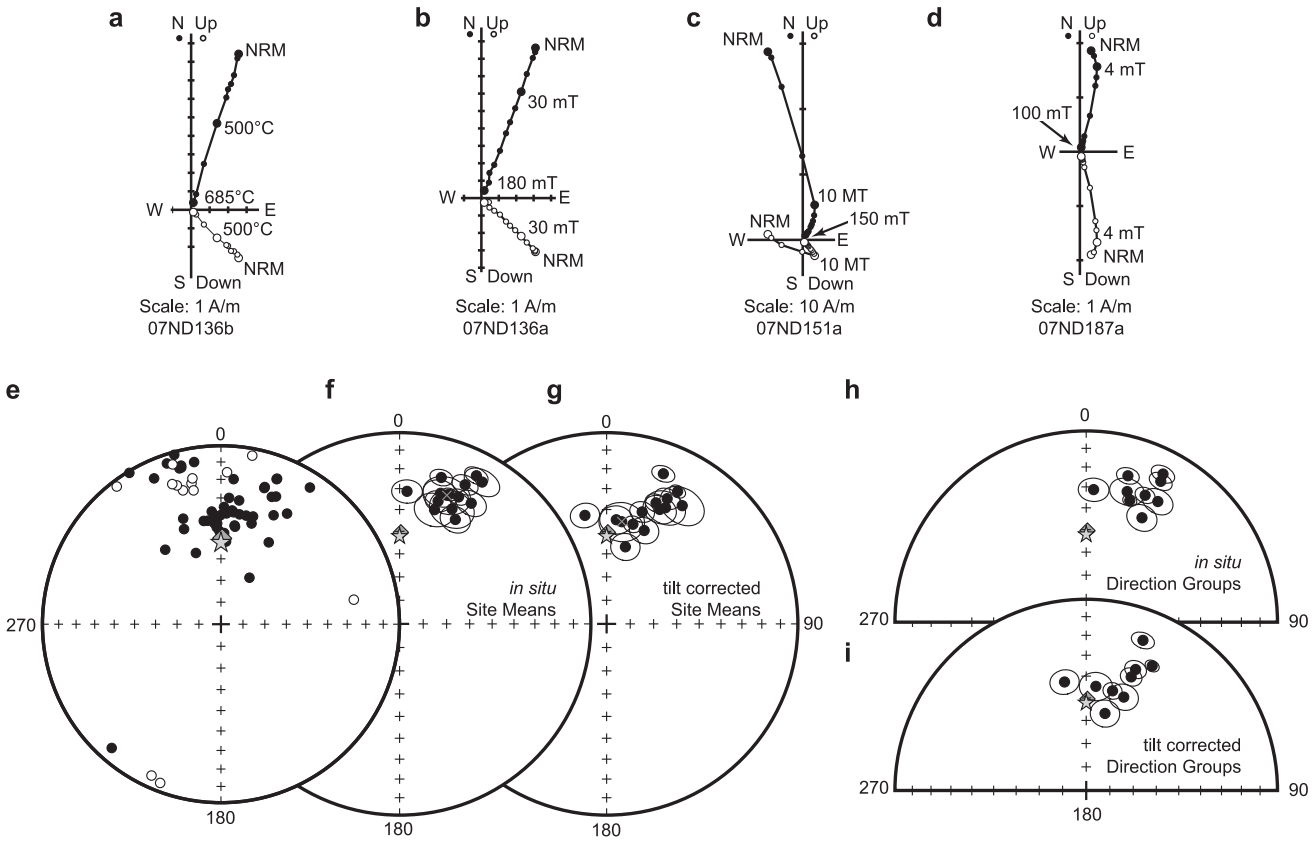


Figure 6. Palaeomagnetic results from Nading Co lavas. Representative thermal (a) and AF (b)–(d) demagnetization results in *in situ* coordinates. (e) Equal area net of low-coercivity overprints. Site mean directions with 95 per cent confidence intervals in *in situ* coordinates (f) and in tilt-corrected coordinates (g). Group directions with 95 per cent confidence intervals in (h) *in situ* coordinates and in (i) tilt-corrected coordinates. Other conventions and symbols are the same as in Figure 4.

Table 2. Maximum likelihood estimates of tilt-corrected Qiangtang palaeolatitudes.

Scenario	N	λ−MLE (°N)	Δλ-MLE (°N)	κ	S (°)
Site means	33	28.9	3.0	41.2	12.6
Direction groups	20	28.7	3.7	45.9	12.0

Notes: Maximum likelihood estimates (λ-MLE) of palaeolatitude. Δλ-MLE is the 95 per cent confidence limit on the maximum likelihood estimate; κ is the best estimate of Fisher's (1953) precision parameter; S the mean dispersion of the palaeolatitude estimate. MLEs and κ-values calculated after Arason & Levi (2010). See text for details.

5.6 Averaging palaeosecular variation

The accuracy of our palaeolatitude calculations rests not only on accurate structural corrections, but also on the ability of the sample set to time-average secular variation sufficiently that a good estimate of the late Palaeogene Earth's magnetic field assuming a GAD field is obtained. A generally accepted approach to evaluate averaging of secular variation is to examine the angular dispersion (S), or statistical scatter, of VGPs in a sample set, assuming that each VGP (or virtual palaeolatitude) is an independent and instantaneous record of the geomagnetic field. The calculated dispersion can then be compared directly to secular variation models (e.g. Tauxe & Kent 2004) and the scatter observed in ancient lavas (e.g. Biggin *et al*. 2008; Johnson *et al.* 2008) that do yield good estimates of the GAD field when averaged.

The angular dispersion of the Qiangtang lava data set can be calculated from the Fisher's precision parameter (K) that is provided by the maximum likelihood estimates (Arason & Levi 2010) using

$$S = (2/K)^{0.5}, \tag{2}$$

where S is in radians (Cox 1970). The direction groups provide independent samples of the ancient geomagnetic field and are therefore most suitable for evaluating secular variation. The Arason & Levi (2010) maximum likelihood method does not calculate the uncertainty on the precision parameter, so we assign an arbitrary and liberal precision uncertainty of ±5; in truth, this value may be higher. Our calculated dispersion of 12.0° (95 per cent confidence interval of 11.4–12.7°) at 29°N palaeolatitude is consistent with the 15.6° (12.5–18.8)° of dispersion observed in VGP populations from young lavas at the same latitude (Tauxe *et al*. 2000; Johnson *et al*. 2008). Thus, the observed virtual palaeolatitude dispersion calculated from Qiangtang lava direction groups is consistent with predicted values and can be straightforwardly explained by secular variation. We conclude that our measured inclinations can be used to reliably evaluate the upper Palaeogene latitude of southern Qiangtang. Only results from tilt-corrected direction group data are considered in the following discussion.

6 DISCUSSION

6.1 Comparisons with reference poles

Palaeolatitudes observed in the upper Palaeogene lavas from southern Qiangtang suggest that Central Tibet was located at 28.7 ± 3.7°N during Late Eocene time. The significance of this palaeolatitude, especially in terms of the Cenozoic tectonics of Asia, is best assessed by comparison with palaeolatitudes for this same location predicted by various palaeomagnetic reference poles and with estimates of shortening within Central Asia from the regional geological record. We take two approaches for selecting reference poles from which to make intra-Asia convergence estimates. First, we use the 40 and 30 Ma poles from the Eurasia synthetic APWP of Torsvik *et al*. (2008), which supersede the widely used reference poles of Besse & Courtillot (2002).

The second approach selects what we consider the palaeomagnetic data from Palaeogene rocks from stable Asia that are most likely to define a robust palaeomagnetic pole and those which are most comparable in sampling strategy and data analysis to our Qiangtang data set. These data come from recent studies by Hankard *et al*. (2007) and Dupont-Nivet *et al*. (2010b). Because post-30 Ma shortening between Mongolia and Siberia is negligible (Cunningham *et al.* 1996; Cunningham 2005; Jolivet *et al.* 2007), palaeomagnetic poles from Mongolia should represent stable Asia. Table 3a shows two Mongolia poles calculated from 39 to 28 Ma flat-lying basalts in southwest Mongolia. The first includes palaeomagnetic data from 33 lava flows from three separate localities (Hankard *et al.* 2007), while the second Mongolian reference pole includes palaeomagnetic directions from 18 additional lavas from three new localities (Dupont-Nivet *et al.* 2010b). The data used for these poles are shown in Table A4, and both poles are the Fisher mean of site VGPs.

We do not compare our results to the recently compiled Cenozoic APWP for the Tarim basin (Gilder *et al*. 2008). These Tarim palaeomagnetic data are from sedimentary rocks and the underlying palaeomagnetic directions from which these poles are calculated have not been evaluated or corrected for possible sedimentary inclination shallowing biases (see Section 6.4.2). Sedimentary inclination shallowing of the Tarim palaeomagnetic directions may explain why the Cenozoic reference pole for Tarim is far-sided with respect to the lava-based Mongolian pole of the same age.

The amount of implied intracontinental convergence between Qiangtang and stable Asia is the difference between the palaeolatitude at a site in southern Qiangtang (at 33°N, 88°E) that would be expected from the reference pole and the palaeomagnetically observed palaeolatitude for the southern Qiangtang volcanics. The statistical significance of the consistency between the southern Qiangtang volcanic data and reference poles is evaluated using a Student's *t*-test as described by Doubrovine & Tarduno (2008). We test the null hypothesis that the two palaeolatitude estimates (the southern Qiangtang lavas and the estimate from a given reference pole) are from the same populations sharing a common mean, that is, H_0: $\lambda_{observed} = \lambda_{predicted}$. The null hypothesis can be rejected (the palaeolatitudes are different) at a specified alpha significance level if $|t'| > t_{alpha/2}(df)$ where $t_{alpha/2}(df)$ is the critical value of the t distribution with df as degrees of freedom. The associated p-value of each test is calculated to assess the lowest significance level at which the null hypothesis is rejected based on the observed value of the t-statistic. The results of these tests are shown in Table 3b.

The Qiangtang palaeolatitudes are different at 99 per cent or greater confidence from the palaeolatitudes predicted from both of the Eurasia reference poles (Table 3b; Fig. 7). In contrast, the observed palaeolatitude for Qiangtang is different from those predicted from the Mongolian poles at only ∼73 per cent or less confidence. In terms of statistical significance, the calculated Qiangtang palaeolatitude is clearly most consistent with predictions from the Mongolian reference pole. Notably, the Mongolia and Eurasia poles are significantly different from each other (Table 3a), so both cannot accurately represent the position of stable Asia. Indeed, the Eurasia reference pole of Torsvik *et al*. (2008) predicts ∼1190 ± 410 km of northward displacement of Mongolia with respect to Siberia since 40 Ma. This motion would have to be accommodated by lithospheric shortening between Mongolia and the stable Asia portion of

Table 3a. Reference poles for Stable Asia used in this study.

Reference pole	Pole age (Ma)	N	λ (°N)	ϕ (°E)	A_{95}	κ	Reference
1. Eurasia	30	18	82.7	152.5	2.8	153.4	Torsvik *et al.* (2008)
2. Eurasia	40	19	82.3	150.5	2.8	144.6	Torsvik *et al.* (2008)
3. Mongolia	33	33	82.5	247.8	4.0	39.9	Hankard *et al*. (2007)
4. Mongolia	33	51	82.9	255.3	3.8	28.7	Dupont-Nivet *et al.* (2010a)

Notes: N, the number of poles in the reference pole; λ (ϕ), latitude (longitude) of reference pole; A_{95}, half angle of the 95 per cent confidence cone of reference pole; κ, the Fisher precision parameter of the mean pole. Data used to calculate the Mongolia poles can be found in Table A4.

Table 3b. Convergence rstimates and consistency tests.

Reference pole	$\lambda_{exp} \pm \Delta\lambda_{exp}$	Latitude displacement	p-Value (per cent)	N–S shortening	Reference
1. Eurasia	35.9 ± 2.3	7.2 ± 4.4	0.2	799 ± 484	Torsvik *et al.* (2008)
2. Eurasia	36.3 ± 2.3	7.6 ± 4.4	0.3	844 ± 484	Torsvik *et al*. (2008)
3. Mongolia	25.9 ± 3.2	−2.8 ± 4.9	26.2	−311 ± 543	Hankard *et al*. (2007)
4. Mongolia	26.1 ± 3.0	−2.6 ± 4.8	28.7	−289 ± 532	Dupont-Nivet *et al*. (2010a)

Notes: Consistency tests for the South Qiangtang data with reference poles listed in Table 3a. λ_{exp}, palaeolatitude in degrees north expected at a reference site (33°N, 88°E) in southern Qiangtang from the reference pole; Latitudinal Displacement is the discrepancy in degrees latitude between the palaeolatitude predicted from the given reference pole and the palaeolatitude calculated from the southern Qiangtang volcanics; p is the probability of observing the value of the associated t-statistic (t') as large as t' provided that the observed and predicted palaeolatitudes are the same; N–S Shortening is the magnitude of northward convergence (in km) between the given reference frame and the reference site in southern Qiangtang.

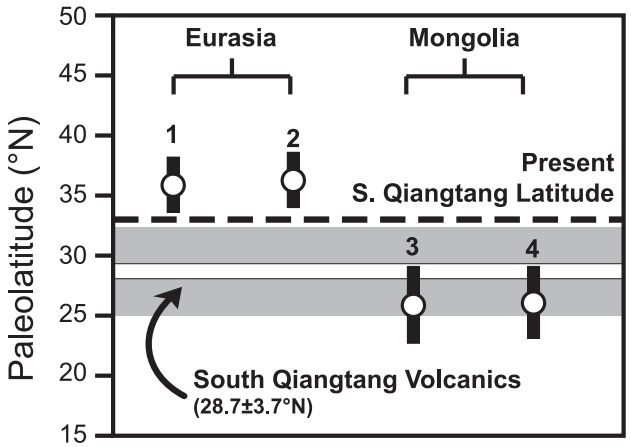


Figure 7. Upper Palaeogene latitudes for a reference site in southern Qiangtang (33°N, 88°E) predicted from selected reference poles. Bold numbers correspond to poles listed in Table 3a. The maximum likelihood estimate of palaeolatitude calculated from the direction groups of the Qiangtang volcanics in tilt-corrected coordinates is shown as the thick white line at 28.7°N latitude, with the 95 per cent confidence limit as the gray box. The palaeolatitude calculations assume a 100 per cent geocentric axial dipole (GAD) geomagnetic field.

Eurasia during Neogene time, which is geologically unrealistic. Thus, as noted previously (e.g. Cogné *et al.* 1999; Hankard *et al.* 2007; Dupont-Nivet *et al.* 2010b), the Cenozoic Eurasia reference poles predict intracontinental convergence values that are irreconcilable with the geological record of upper crustal shortening in Asia.

6.2 Uncertainty in the reference poles

The large discrepancies between the Mongolia and Eurasia poles taken to represent a stable Asia reference frame indicate significant inaccuracies, and therefore limitations, in one or both of these poles for predicting Cenozoic palaeolatitudes in Asia. The Mongolia poles (Hankard *et al.* 2007; Dupont-Nivet *et al.* 2010b) are derived from well-dated lavas from stable Asia that have good structural control and which record the long-term behaviour of the geomagnetic field. Therefore, like Cogné *et al*. (1999), Hankard *et al*. (2007), and Dupont-Nivet *et al*. (2010b), we conclude that the greatest uncertainties lie in the use of the Eurasia reference poles to represent stable Asia.

There are several potential sources of error in the Eurasia poles that would compromise their utility as a stable Asia reference frame, many of which are also discussed in Dupont-Nivet *et al*. (2010b). First, inaccuracies in the global plate circuits used to compile and then transfer the synthetic reference poles into various continental reference frames may introduce modest errors, perhaps up to a few hundreds of kilometres, in the estimated position of continents (Royer *et al.* 2006; Horner-Johnson *et al.* 2007). Consistency between palaeomagnetic data from Upper Cretaceous to lower Palaeogene volcanics from North American and Pacific hotspots transferred into North American coordinates suggest that plate circuit inaccuracies are small (Doubrovine & Tarduno 2008), and therefore do not appear sufficient to account for most of the discrepancy between the palaeolatitudes predicted from the synthetic APWPs and those observed in lavas from Asia.

Another possible source of error in the Eurasia APWP poles that could make them unsuitable for Cenozoic palaeogeographic reconstruction of Asia is the inclusion of spurious or questionable palaeomagnetic data in the synthetic reference pole compilations. For example, many APWP compilations include palaeomagnetic poles determined from continental sediments, which are especially prone to inclination shallowing biases (see Section 6.4.2). Even some of the most thorough and modern palaeomagnetic studies of lava sequences such as several within the North Atlantic Igneous Province yield inconsistent results, indicating as yet unresolved complications with palaeomagnetic data from lower Palaeogene volcanics often used in global reconstructions (Riisager *et al.* 2002, 2003; Ganerød *et al.* 2008; Doubrovine & Tarduno 2008). Importantly, although the individual poles used in the 40 and 30 Ma synthetic reference poles of Torsvik *et al*. (2008) appear to be of high quality and reliability, the small number of these poles averaged over a long (20 Ma) time window and the non-uniform distribution of the sampling locations of these poles probably introduces significant error in the position of the average reference poles. We note that the 40 and 30 Ma synthetic APWPs of Torsvik *et al*. (2008) contain only 2 and 1 poles, respectively, from Eurasia, and none of these are from Asia.

Third, non-rigid motion between Europe and Asia would invalidate the assumption that Europe and Asia have moved as a large, coherent tectonic unit throughout the Cenozoic, and therefore make the Eurasia APWPs unsuitable for predicting Asia palaeolatitudes (Cogné *et al.* 1999). Although deformation along the Tornquist-Tesseyre line or within the Ural mountains of eastern Europe has been suggested as potential loci of Palaeogene decoupling between Europe and Asia (Cogné *et al.* 1999; Hankard *et al.* 2007), the geological records of these areas suggest these lineaments have been tectonically quiescent throughout the Cenozoic. Furthermore, detailed reconstructions of Arctic Ocean demonstrate that the Eurasian Basin has been in extension since ∼55 Ma (Alvey *et al*. 2008). Thus, large-scale decoupling between Europe and Asia appears unlikely.

In summary, there are several potential sources of uncertainty and inaccuracy associated with using Eurasia reference poles to predict the Cenozoic palaeogeography of Asia. Well-dated and well-determined lava-based palaeomagnetic poles from stable Asia avoid the limitations of all of them. We concur with Hankard *et al*. (2007) and Dupont-Nivet *et al*. (2010b) that these stable Asia reference poles are most suitable and accurate for predicting the palaeogeography of Asia and are therefore the preferred reference data against which to compare palaeomagnetic data from the tectonically active regions of Asia.

6.3 Implications for the Central Asia inclination anomaly

The Central Asia inclination anomaly refers to the well-established observation that most magnetic inclinations measured in Cenozoic rocks from Asia are much shallower than those predicted from the coeval Eurasia reference poles (e.g. Westphal 1993; Chauvin *et al.* 1996; Cogné *et al.* 1999; Hankard *et al.* 2007). Because Eurasia reference poles have traditionally been assumed to accurately represent the palaeogeography of stable Asia, the inclination anomaly has been interpreted to indicate 1000s of km of post-60 Ma convergence within Asia, non-rigid motion between Asia and Europe (Cogné *et al.* 1999), the occurrence of large local geomagnetic anomalies or zonal non-dipole fields that bias the expected GAD field directions (e.g. Chauvin *et al.* 1996; Van der Voo & Torsvik 2001; Si & Van der Voo 2001), or sedimentary processes that bias remanent magnetic directions toward shallower inclinations and lower palaeolatitudes (e.g. Gilder *et al.* 2001; Tan *et al.* 2003; Tauxe & Kent 2004; Tauxe 2005). The magnitude of the inclination anomaly

decreases significantly, however, when only the highest quality lava-based palaeomagnetic data from Central Tibet and stable Asia are considered in tectonic reconstructions (Table 3b and Fig. 7). This observation suggests that rocks from Central Tibet and stable Asia record a similar magnitude of inclination shallowing, regardless of the process responsible for that shallowing. We have previously discussed the inability of tectonics, either as large-magnitude intracontinental shortening or non-rigid behaviour of Asia, to explain the inclination anomaly. Below we evaluate the non-dipole hypothesis and inclination shallowing hypothesis for the origin of the Central Asia inclination anomaly in the context of volcanic-based palaeomagnetic data from Central Tibet and stable Asia.

6.4.1 Non-dipole fields

Several authors have suggested that the Central Asia inclination anomaly arises from a persistent local geomagnetic anomaly (Chauvin *et al.* 1996) or global axial quadrupolar (G_2) or octupolar (G_3) components to the geomagnetic field (Si & Van der Voo 2001; Van der Voo & Torsvik 2001) that biases primary remanent directions toward inclinations more shallow than predicted from a GAD field. Dupont-Nivet *et al*. (2010b) tested the non-dipole hypothesis by searching for the axial octupolar field contribution that minimizes the discrepancy between the palaeomagnetic data from lavas and inclination-corrected sediments in Asia and the Eurasia synthetic reference poles of Torsvik *et al.* (2008). They found that the inclination anomaly is minimized when the axial octupolar component is ∼4 per cent during the Miocene (20 Ma), 14–16 per cent during the Middle Eocene (40 Ma) and ∼8 per cent during the Cretaceous (∼120 Ma). Thus, Dupont-Nivet *et al*. (2010b) concluded that a persistent axial octupolar component to the GAD field can fully explain the inclination anomaly, but it would have to be greater than a few per cent and vary in magnitude throughout the Cenozoic.

Although the discrepancies among the 30–40 Ma palaeomagnetic poles can be largely accounted for by assuming G_3 field components up to 14 per cent, the hypothetical non-dipole field solution is non-unique and could be composed of a varying combination of quadrupolar, octupolar and higher order components that would also produce the observed discrepancies. The present distribution of high-quality palaeomagnetic data in space and time is insufficient to rigorously evaluate non-dipole geomagnetic field effects (Merrill *et al*. 1996; McElhinny 2004). Studies of the much better documented 0–5 Ma geomagnetic field, however, suggest that G_2 and G_3 terms contributed only a few per cent to the time-averaged geomagnetic field (Quidelleur *et al.* 1994; Merrill *et al.* 1996; McElhinny & McFadden 1997; Johnson *et al.* 2008). Thus, if the 0–5 Ma geomagnetic field is characteristic of the Cenozoic geomagnetic field, then the GAD hypothesis is sufficiently accurate for purposes of tectonic reconstruction. A small (5 per cent), persistent octupole component may improve these reconstructions (Dupont-Nivet *et al*. 2010b).

Nonetheless, if an axial G_3 field component of dominantly one sign and of greater than 10 per cent of the geomagnetic field did persist during late Palaeogene time, then intracontinental convergence estimates calculated assuming a GAD field should be reevaluated. Observed palaeolatitudes can be adjusted for such a zonal octupole contribution using the following equation (Merrill *et al.* 1996):

$$\tan(\lambda_{obs}) = \tan(\lambda)\{[2 + 2G_3(5\sin^2\lambda - 3)]/[2 + 3G_3(5\sin^2\lambda - 1)]\}, \quad (3)$$

where λ_{obs} is the palaeolatitude obtained from the palaeomagnetically observed inclination I_{obs} using the standard formula for a GAD field (eq. 1), λ is the corrected palaeolatitude assuming that the time-averaged field actually did include a persistent octupolar component and G_3 is that octupole component expressed as a percentage of the geomagnetic field. Table 4 shows palaeolatitudes calculated from the directions observed in the lavas from Qiangtang and Mongolia corrected to account for a 14 per cent axial G_3 component (Dupont-Nivet *et al*. 2010b). Although a significant G_3 component does lower palaeolatitudes observed in the Asian lavas by ∼9°, the *relative* latitudinal separation between Qiangtang and Mongolia does not change significantly (Table 4) because the N–S separation is small compared to the scale of the octupolar anomaly. Thus, although very large-scale non-dipole fields greater than a few per cent significantly distort the time-averaged field from GAD (Dupont-Nivet *et al*. 2010b), reliable intracontinental convergence estimates can be maintained by restricting our analysis to reference poles from Asia that are relatively close to Qiangtang.

Table 4. Octupole-corrected palaeolatitudes for Central Asia.

Terrane	GAD+14 per cent G_3 λ_{obs} (°N) ± $\Delta\lambda_{obs}$ (°)	GAD λ_{obs} (°N) ± $\Delta\lambda_{obs}$ (°)
Qiangtang	28.7 ± 3.7	37.6 ± 4.6
Mongolia	38.5 ± 3.0	48.3 ± 3.1
Latitudinal separation	9.8 ± 4.8	10.7 ± 5.5

Notes: Observed palaeolatitudes are calculated from the palaeomagnetic directions recorded in the lavas from Qiangtang and Mongolia and here are assumed to record a GAD+14 per cent G_3 field component. Corrected palaeolatitudes have removed the shallow bias imparted by the G_3 component. Observed Mongolia data from Dupont-Nivet *et al.* (2010a). See text for details.

6.4.2 Sedimentary processes and inclination shallowing

The Central Asia inclination anomaly has been largely framed in the context of palaeomagnetic results from the sedimentary record (e.g. Enkin *et al.* 1992; Cogné *et al.* 1999; Gilder *et al.* 2008). Syndepositional and post-depositional processes such as particle settling or compaction can shallow primary remanent directions recorded in sedimentary rocks (King 1955; Verosub 1977; Anson & Kodama 1987), yielding palaeolatitudes that are too low if not detected and corrected. Volcanic rocks are immune to sedimentary flattening processes, and therefore provide the most reliable palaeolatitude estimates if primary thermoremanent directions are preserved and palaeosecular variation is sufficiently time-averaged in the sampled lava section. Therefore, robust volcanic-based palaeomagnetic data provide a powerful test for the occurrence and magnitude of sedimentary inclination shallowing.

We assess the magnitude of sedimentary inclination shallowing in Central Tibet by comparing our volcanic data to sedimentary-based palaeomagnetic data from three studies of Palaeogene sediments from the Hoh Xil basin in the Qiangtang terrane (Table A5, Lin & Watts 1988; Cogné *et al.* 1999; Liu *et al.* 2003). The expected inclination for a reference site at 33°N, 88°E was calculated from the mean VGP of each sampling locality (I_{obs}, Table 5). To assess the magnitude of inclination shallowing in the Qiangtang sediments, I_{obs} was compared to the maximum likelihood estimate of inclination of the Qiangtang lavas (I_{cor}), and the discrepancy between lava and sedimentary data is referred to as the 'correction' (Cor ± ΔCor). The mean correction calculated from Qiangtang sediments is skewed toward higher values by the results of Lin & Watts (1988), which have much shallower inclinations than those observed in coeval sections from the same area reported by Liu *et al*. (2003)

Table 5. Sedimentary inclination shallowing in cenozoic Central Asian Redbeds.

		Site location		Observed Inc.	Reference pole				Expected Inc.	Corrected Inc.	Correction	
Formation	Age (Ma)	Lat. (°N)	Long. (°E)	$Inc_{obs} + \Delta I_{obs}$	Age (Ma)	Lat. (°N)	Long. (°E)	A_{95}	$I_{exp} \pm \Delta I_{exp}$	$I_{cor} \pm \Delta I_{cor}$	Cor + ΔCor	Reference
Potwar	10	33.3	72.3	34 ±4	10	87.2	125.0	2.5	54 ± 2	41 ± 8	7 ± 9	Tauxe (2005)
Guide basin	2.6–11.5	36.1	101.5	44 ±2	10	87.2	125.0	2.5	58 ± 2	57 ± 5	13 ± 5	Yan *et al.* (2005)
Subei	9–20	39.5	94.7	44 ±2	10	87.2	125.0	2.5	61 ± 2	63 ± 7	19 ± 7	Tauxe (2005)
Chetougou	17–23	36.5	101.9	45 ±4	20	85.0	137.4	3.0	60 ± 2	52 ± 9	7 ± 10	Dupont-Nivet *et al.* (2010a)
Xiejia	23–30	36.5	101.9	50 ±3	20	85.0	137.4	3.0	60 ± 2	53 ± 5	3 ± 6	Dupont-Nivet *et al.* (2010a)
*FHS-GG	30–31	33.0	88.0	35 ±8	30	82.7	153.5	2.8	55 ± 2	48 ± 4	13 ± 9	Liu *et al.* (2003)
*FHS-SP	32–35	33.0	88.0	35 ±10	30	82.7	153.5	2.8	55 ± 2	48 ± 4	12 ± 11	Liu *et al.* (2003)
*FHS-GS	35–40	33.0	88.0	38 ±10	30	82.7	153.5	2.8	55 ± 2	48 ± 4	10 ± 11	Liu *et al.* (2003)
*FHS-LW	40	33.0	88.0	15 ±14	40	82.3	150.5	2.8	56 ± 2	48 ± 4	32 ± 14	Lin & Watts (1988)
Mahalagou	30–41.5	36.5	101.9	43 ±3	40	82.3	150.5	2.8	60 ± 2	54 ± 5	11 ±6	Dupont-Nivet *et al.* (2010a)
*Xialaxiu	50	33.0	88.0	38 ±9	50	79.1	154.2	2.6	56 ± 2	48 ± 4	9 ± 10	Cogné *et al.* (1999)

Notes: Inc_{obs} (ΔInc_{obs}), observed inclination (95 per cent confidence limit on the observed inclination); I_{exp} (ΔI_{exp}), inclination expected at the sampling site from the coeval Eurasian reference pole of Torsvik *et al.* (2008) (95 per cent confidence of expected inclination); I_{cor} (ΔI_{cor}), inclination corrected for sedimentary shallowing using the E/I method of Tauxe & Kent (2004), except those studies marked with *, in which the correct inclination is assumed to be the inclination observed in coeval volcanic rocks (95 per cent confidence of corrected inclination); Cor (ΔCor): the magnitude of the inclination correction, or the difference between the corrected and observed inclination (95 per cent confidence of the correction value).

(Fig. 8 and Table 5). This large difference in the directional results from coeval sedimentary sections may be analytical, or it could be real due to differences in magnetic mineralogy and sedimentary grain size (Dupont-Nivet *et al.* 2002a,b). The mean discrepancy between magnetic inclination recorded in Palaeogene sediments and lavas from Qiangtang is $\sim 15 \pm 5^\circ$. If the results of Lin & Watts (1988) are excluded, then this discrepancy is reduced to $11 \pm 5^\circ$.

If a sedimentary palaeomagnetic data set is large enough and free of secondary overprints, then the inclination flattening bias can be evaluated independently of comparisons to coeval lava data by examining the shape of the distribution of magnetic directions. The shallowing bias often follows the inclination error formula first described by King (1955):

$$\tan(I_{obs}) = f \tan(I_A), \tag{4}$$

where I_{obs} is the observed inclination, I_A the applied field inclination and f is a flattening factor between 0 and 1. Eq. (4) indicates that if an approximately Fisherian distribution of magnetic directions is flattened, then inclination values will become more shallow and the overall distribution will elongate in the horizontal dimension. If the shape of an observed distribution of magnetic directions is consistent with flattening, then it can be 'corrected' using the Elongation/Inclination (E/I) method to find the flattening factor (f in eq. 4) that returns the direction distribution expected for palaeomagnetic secular variation at the sampling locality (Tauxe & Kent 2004; Tauxe 2005). The observed palaeomagnetic data are corrected by inverting eq. (4)

$$\tan(I_{cor}) = (1/f_{est}) \tan(I_{obs}), \tag{5}$$

where I_{obs} is the observed inclination, f_{est} is the estimated flattening factor based on the E/I method and I_{cor} is 'corrected' inclination. We are unable to use the E/I method to evaluate the magnitude of flattening in sedimentary rocks from Qiangtang because individual sample directions are either unavailable, too few for E/I analysis, or much more dispersed than can be attributed to full-polarity secular variation. Several large magnetostratigraphic data sets from the Xining and Lanzhou regions of Northeastern Tibet show a characteristic horizontal elongation consistent with significant flattening biases, however, and have been 'corrected' using the E/I method (Table 5, Tauxe 2005; Yan *et al.* 2005; Dupont-Nivet *et al.* 2010b). The mean inclination correction calculated from these E/I studies is $\sim 10 \pm 3^\circ$.

The magnitude of inclination corrections for Palaeogene sediments from Qiangtang calculated by direct comparison to coeval lavas is remarkably similar to the inclination corrections for sedimentary sections from Northeastern Tibet determined using the E/I method (Table 5). We consider the direct comparison of coeval volcanic and sedimentary rocks to be a more robust inclination correction method, as it is based on field observations in contrast to best-fitting models of direction distributions and assumptions about geomagnetic field behaviour. The consistency in the magnitude of these inclination corrections determined using independent methods—direct comparison of coeval volcanic and sedimentary rocks and palaeosecular variation modeling—may be fortuitous, but is nonetheless encouraging.

This analysis suggests that sedimentary processes are responsible for $\sim 15 \pm 5^\circ$ and $\sim 10 \pm 3^\circ$ of inclination shallowing observed in Palaeogene sediments from Qiangtang and Northeastern Tibet, respectively. These regionally mean values are not blanket corrections, however, because the magnitude of sedimentary inclination shallowing is highly variable and must be evaluated on a case-by-case basis (e.g. Cor ± ΔCor in Table 5). Nonetheless, we

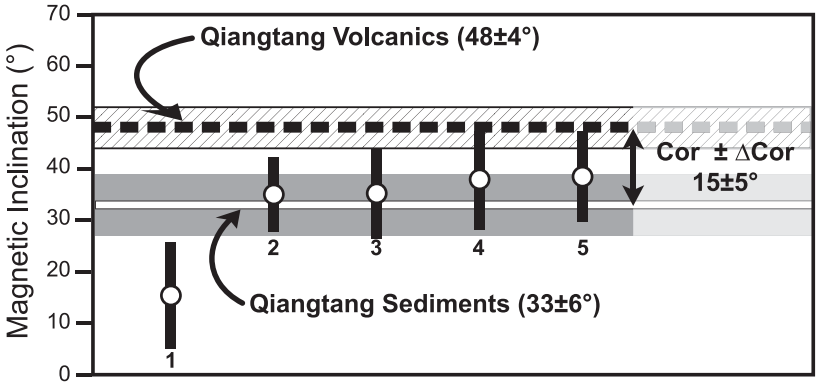


Figure 8. Comparison of palaeomagnetic data from Qiangtang volcanics and sediments. Maximum likelihood estimate (MLE) of inclination from volcanic direction groups shown as thick black dashed line at 48° with 95 per cent confidence level as diagonal-ruled box. The mean inclination discrepancy (Cor ± ΔCor, see Table 5) between Palaeogene sediments and lavas from Qiangtang is ∼15 ± 5°, resulting in a mean inclination of Qiangtang sediments of 33 ± 6°. If the results of Lin & Watts (1988) are excluded, these values change to 11 ± 5° and 37 ± 6°, respectively. The Qiangtang sedimentary data shown are: (1) Fenghuo Shan (Lin & Watts 1988); (2) Fenghuo Shan Section GG (Liu *et al.* 2003); (3) Fenghuo Shan Section SP (Liu *et al.* 2003); (4) Fenghuo Shan Section GS (Liu *et al.* 2003) and (5) Xialaxiu (Cogné *et al.* 1999). See text and Table 5 for details.

emphasize the following two observations in the context of using the Mongolian poles as a stable Asia reference frame. First, sedimentary inclination shallowing on the order of ∼10° can explain why the Tarim reference poles, which are based predominantly on uncorrected palaeomagnetic data from sedimentary rocks (Gilder *et al.* 2008), predict palaeolatitudes that are lower than those observed in upper Palaeogene lavas from Qiangtang and Mongolia. The Tarim reference poles should be used with caution when predicting the Cenozoic palaeogeography of Asia until the underlying palaeomagnetic data have been fully evaluated for sedimentary inclination shallowing (e.g. Tan *et al.* 2003). Second, an ∼10–15° inclination correction applied to most sedimentary-based palaeomagnetic data sets from Central Asia reduces the discrepancy between inclinations predicted from the Mongolian reference poles and the corrected sedimentary directions to a magnitude that is consistent with regional upper-crustal shortening estimates (see below). Thus, the Central Asia inclination anomaly, as observed in palaeomagnetic results from sediments, appears to arise primarily from the choice of the reference pole against which to compare Asia palaeomagnetic data and secondarily from modest sedimentary shallowing biases and Cenozoic shortening on the order of a few hundreds of km within Central Asia.

6.5 Implications for the Palaeogene tectonics of Tibet

Most stratigraphic, metamorphic and palaeomagnetic records suggest the India–Asia collision, defined here as the accretion of the Tethyan Himalaya to the southern margin of Asia, began ∼55 Ma or shortly thereafter at ∼20°N latitude (Patriat & Achache 1984; Rowley 1996; Dupont-Nivet *et al*. 2010a; Lippert *et al*. 2010; Najman *et al*. 2010). Plate reconstructions suggest that there have been ∼3000 km of N–S convergence between India and Asia at the longitude of Lhasa city since 50 Ma (Copley *et al.* 2010), so a key question is where is this convergence partitioned in these plates?

Palaeomagnetically determined N–S convergence estimates (Table 3b), in principal, can be used to calculate the amount of lithospheric shortening or extension between the Lhasa terrane, Qiangtang, and stable Asia since late Palaeogene time. Comparing our calculated palaeolatitude for southern Qiangtang to the palaeolatitudes predicted from upper Palaeogene lavas in Mongolia suggests −2.6 ± 4.8°, or −289 ± 532 km of N–S directed shortening. Asia has been in a state of N–S contraction throughout the Cenozoic, so this palaeomagnetic estimate really translates to 0 to 243 km of N–S directed post- ∼35 Ma upper crustal shortening between Qiangtang and Mongolia.

Structural studies provide independent estimates of upper crustal shortening and extension and therefore place first-order constraints on the plausibility of this palaeomagnetically determined convergence estimate. Minimum N–S shortening estimates for thrust belts between southern Qiangtang and Mongolia since the late Palaeogene are ∼360 km, or 3–4° of latitude, as shown in Fig. 9 and Table 6 (see references therein). The Qiangtang–Mongolia comparison appears to underpredict the magnitude of latitudinal convergence between these terranes as revealed by geological studies, possibly drawing into question the ability of these palaeomagnetic data to accurately predict Tibetan palaeolatitudes. As discussed above (and in Dupont-Nivet *et al.* 2010b), however, volcanic-based palaeomagnetic data that average secular variation and the few sedimentary-based data from Asia that have been evaluated and corrected for inclination shallowing should provide the most robust palaeolatitude estimates. Perhaps the uncertainties in the palaeomagnetic data have not been fully characterized, which is testable with additional sampling of Palaeogene lavas in Tibet and Mongolia.

Our palaeolatitude estimate for Southern Qiangtang is consistent with recent palaeomagnetic work to determine the upper Cretaceous and lower Palaeogene latitude of the Lhasa block and the Cenozoic shortening record of the Lhasa block. A compilation of robust volcanic-based palaeomagnetic data from the 56 to 47 Ma upper Linzizong formation near Lhasa indicates that the southern margin of Tibet at the longitude of Lhasa was 20 ± 3°N at ∼50 Ma (Achache *et al*. 1984; Lin & Watts 1988; Chen *et al*. 2010; Dupont-Nivet *et al*. 2010a; Liebke *et al*. 2010; Sun *et al.* 2010; Tan *et al*. 2010, as summarized in Lippert *et al*. 2010). The 50 Ma latitude of Lhasa is remarkably consistent with the 22 ± 2°N Late Cretaceous latitude of the same region calculated from inclination-shallowing corrected red beds of the upper Cretaceous Takena formation (Tan *et al*. 2010), suggesting that the latitude of the Lhasa block remained relatively stationary prior to the collision with the Tethyan Himalaya. If we assume that the latitude of the Lhasa block remained just as stable between 50 and ∼35 Ma, then we can predict the ∼35 Ma latitude of Southern Qiangtang using the Palaeogene shortening record and 50 Ma latitude of the Lhasa block. Post-50 Ma

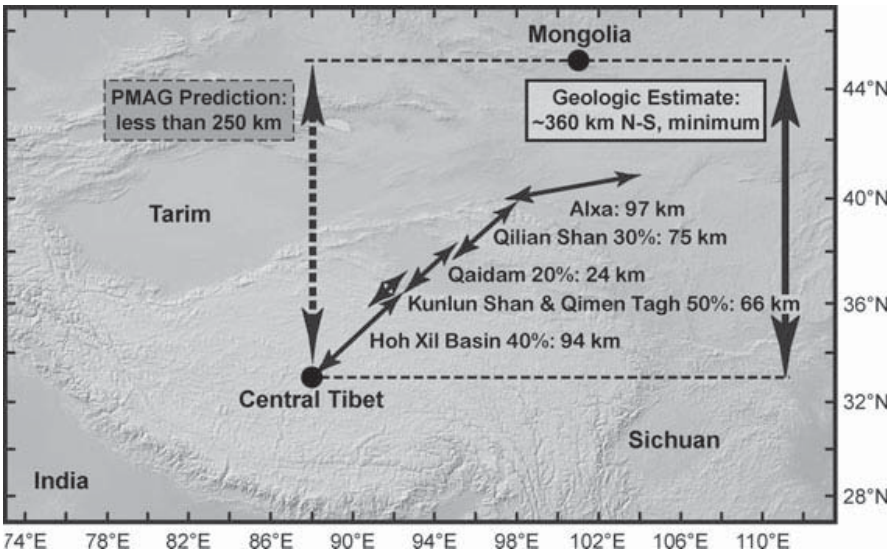


Figure 9. Comparison of post-40 Ma north–south-directed intra-Asia convergence estimates as calculated from volcanic-based palaeomagnetic data from Qiangtang (Central Tibet) and Mongolia and minimum upper crustal shortening estimates as calculated from structural studies. Numbers presented are only the N–S component of upper crustal shortening. See Table 6 for details and sources of these shortening estimates.

Table 6. Minimum post-50 Ma shortening estimates for Central Asia at the Longitude of Lhasa.

Terrane	Shortening (per cent)	Shortening azimuth (E°)	Approx. present width (km)	Pre-Oligocene width (km)	Shortening (km)	N–S shortening (km)	Reference
Bangong Suture Zone	20	0	50	62.5	13	13	Kapp *et al.* (2007a)
Hoh Xil Basin	40	45	200	333	133	94	Coward *et al.* (1988); Wang *et al*. (2008)
Kunlun and Qimen Tagh	50	55	115	230	115	66	Yin *et al*. (2007)
Qaidam	20	55	170	213	43	24	Zhou *et al*. (2006)
Qilian Shan	30	60	350	500	150	75	Brian Darby (personal commmunication, 2008)
Alxa/Altyn Tagh Fault[a]	—	75	375	—	—	97	Yue *et al.* (2005); Darby *et al*. (2005)

Notes: Minimum post-50 Ma shortening estimates for Central Asia at the Longitude of Lhasa-Central Tibet-Mongolia from structural studies.
[a]Shortening in Alxa is actually ∼375 km of northeastward translation of the Gobi region along a northern extension of the Altyn Tagh Fault.

N–S directed upper crustal shortening within southern Tibet at the longitude of Lhasa is less than 100 km (Kapp *et al*. 2005, 2007a,b). Therefore, we assume that the N–S distance between the southern margin of the Lhasa terrane (29°N, 88°E) and our reference site in southern Qiangtang (33°N, 88°E) is 4 ± 1° of latitude. Thus, we predict that the ∼35 Ma latitude for Southern Qiangtang was ∼24 ± 3°N, which is consistent with the palaeolatitude calculated from the upper Palaeogene southern Qiangtang volcanics (28.7 ± 3.7) at 95 per cent confidence (Doubrovine & Tarduno 2008). Modest northward drift of Tibet from 50 to 35 Ma would bring this palaeolatitude prediction into even greater correspondence with the southern Qiangtang volcanics.

In summary, structural shortening budgets and volcanic-based palaeolatitude estimates from Southern Tibet, Central Tibet and Mongolia described above suggest that post-50 Ma shortening north of the Indus-Yarlung suture separating the Tethyan Himalaya from Asia is on the order of a few hundreds up to ∼650 km (Fig. 9, Table 6, see also Dupont-Nivet *et al*. 2010a, van Hinsbergen *et al.* submitted). Like Dupont-Nivet *et al.* (2010a), we conclude that only approximately one third of the post-50 Ma India–Asia appears to be partitioned in Asian lithosphere. The remaining convergence must have been accommodated south of the Indus-Yarlung suture, possibly as a combination of shortening within the Tethyan Himalaya (Ratschbacher *et al*. 1994; Ding *et al.* 2005), subduction of the Indian Plate without accretion of upper crustal units prior to ∼30 Ma (Dupont-Nivet *et al*. 2010a), and shortening within the Himalaya after ∼30 Ma (DeCelles *et al*. 2002). These lower intra-Asia shortening estimates also question the role of eastward lateral extrusion of coherent tectonic blocks out of Central Tibet in accommodating large amounts of India–Asia convergence (Tapponnier *et al*. 1982; Replumaz *et al*. 2004).

The 20–28°N Palaeogene latitudes of Tibet as interpreted from the Lhasa and Qiangtang volcanic-based palaeomagnetic data also have important implications for weathering and erosion of the Tibetan plateau, Cenozoic climate dynamics, and the application of palaeoaltimetry proxies on the plateau and its forelands. Several recent studies suggest that a substantial continental plateau existed on the southern margin of Asia prior to 50 Ma and throughout early Palaeogene time (England & Searle 1986; Murphy *et al.* 1997; Kapp *et al.* 2005, 2007b; Leier *et al.* 2007; Dupont-Nivet *et al.* 2008; Wang *et al.* 2008). The palaeomagnetic data presented here indicate that this proto-Tibetan plateau would have been located at tropical to low subtropical latitudes throughout most of the warm Palaeogene (Zachos *et al.* 2001; Bohaty & Zachos 2003). Thus, the southern margin of the plateau is expected to have been affected by high rates of chemical and physical weathering and organic

carbon burial, much like the Himalaya are today, and the plateau itself would have been positioned in the evaporative and arid subtropics (Richter *et al.* 1992; France-Lanord & Derry 1997; Kent & Muttoni 2008). These palaeogeographic and climatic conditions may have facilitated the development of the high elevation, internal drainage and low relief that characterizes the modern plateau (Fielding *et al.* 1994; Sobel *et al.* 2003; Dupont-Nivet *et al.* 2007; DeCelles *et al*. 2007a,b). Furthermore, the warmer and more evaporative environmental conditions expected in the subtropics must be considered when evaluating stable isotopic and palaeobotanical methods for estimating the palaeoelevation of the plateau (Rowley & Garzione 2007; Ehlers & Poulsen 2009; Galewsky 2009; Insel *et al.* 2009). Finally, regional and global climate models that explore potential links between climate, environmental change and the development of the Tibetan plateau (e.g. Ramstein *et al.* 1997; An *et al.* 2001) should incorporate the revised estimates for the latitude of the Palaeogene Tibetan plateau.

7 CONCLUSIONS

We have carried out a palaeomagnetic study of upper Palaeogene lavas from the Qiangtang terrane in Central Tibet to better understand the cause of the Central Asia inclination anomaly, determine the Palaeogene latitude of Tibet, and refine palaeomagnetic-based intracontinental shortening estimates for Asia. This study presents the first palaeomagnetic results from Cenozoic volcanics from the Qiangtang terrane in Central Tibet. Our conclusions can be summarized as follows:

1. Our $^{40}Ar/^{39}Ar$ geochronologic results from WWH lavas indicate that these volcanics were erupted 38.6 $\pm$ 0.5 Ma. These ages are similar to other basalts in this sampling region and confirm that this sampling locality is part of the east–west belt of upper Eocene potassic volcanic rocks north of the Tanggula Shan and not a younger belt of Hoh Xil lavas.

2. Thirty-eight sampling sites from three localities define twenty direction groups representing independent samples of the geomagnetic field from 30 to 40 Ma. Rock magnetism, demagnetization behaviour and direction distributions all support the conclusion that the ChRMs are primary thermoremanent magnetizations and do not have a strong anisotropy that could bias the observed directions.

3. The maximum likelihood estimate of tilt-corrected palaeolatitudes for the 20 direction groups is 28.7 $\pm$ 3.7°N, confirming that Central Asia was located ∼8–10° farther south in the late Palaeogene than predicted from Eurasia APWPs.

4. Well-dated and well-determined lava-based palaeomagnetic poles from stable Asia are most suitable and accurate for predicting the palaeogeography of Asia and are therefore the preferred reference data against which to compare palaeomagnetic data from the tectonically active regions of Asia. These poles minimize uncertainties introduced by undetected displacements between the reference poles and study sites, as well as uncertainties due to inclination biasing effects of possible non-dipole fields.

5. Magnetic inclinations observed in Cenozoic sedimentary rocks from Asia are lower than those observed in coeval lavas from the same region, but the discrepancy is consistent with the magnitude of sedimentary inclination shallowing inferred from Elongation/Inclination correction methods. Our analysis of sedimentary inclination corrections indicates that sedimentary processes are responsible for ∼10–15° of inclination shallowing in Central Asia.

6. The Central Asia inclination anomaly, the discrepancy between inclinations predicted from Eurasia reference poles and those observed in Cenozoic rocks from Asia, appears to arise primarily from the choice of the reference frame against which to compare Asia palaeomagnetic data and secondarily from Cenozoic shortening on the order of a few hundreds of kilometres within Central Asia and the use of sedimentary-based magnetic directions that have not been corrected for inclination shallowing.

7. A comparison of the southern Qiangtang palaeolatitude reported here with the palaeolatitude calculated from coeval lavas in Mongolia indicates that post- ∼35 Ma intra-Asia convergence is on the order of a few hundred kilometres, consistent with structurally based shortening estimates for the same region and time period of 360–650 km.

8. Southerly, tropical to subtropical Asia palaeolatitudes have important implications for the interpretation of palaeoaltimetry proxies, weathering and erosion of the Tibetan plateau and early Cenozoic climate dynamics.

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REFERENCES

Achache, J., Courtillot, V. & Zhou, Y.X., 1984. Paleogeographic and tectonic evolution of southern Tibet since middle Cretaceous time: new paleomagnetic data and synthesis, *J. geophys. Res.,* **89,** 311–339.

Aitchison, J.C., Ali, J.R. & Davis, A.M., 2007. When and where did India and Asia collide? *J. geophys. Res.,* **112,** doi:10.1029/2006JB004706.

Alvey, A., Gaina, C., Kusznir, N.J. & Torsvik, T.H., 2008. Integrated crustal thickness mapping and plate reconstructions for the high Arctic, *Earth planet. Sci. Lett.,*, **274,** 310–321, doi:10.1016/j.epsl.2008.07.036.

An, Z.S., Kutzbach, J.E., Prell, W.L. & Porter, S.C., 2001. Evolution of Asian monsoons and phased uplift of the Himalaya-Tibetan plateau since Late Miocene times, *Nature,* **411,** 62–66, doi:10.1038/35075035.

Anson, G.L. & Kodama, K.P., 1987. Compaction-induced shallowing of the post-depositional remanent magnetization in a synthetic sediment, *Geophys. J. R. astr. Soc.,* **88,** 673–692.

Arason, P. & Levi, S., 2010. Maximum likelihood solution for inclination-only data in paleomagnetism, *Geophys. J. Int.,* **182,** 753–771, doi:10.1111/j.1365-246X.2010.04671.x.

Baksi, A.K., Archibald, D.A. & Farrar, E., 1996. Intercalibration of $^{40}Ar/^{39}Ar$ dating standards, *Chem. Geol.,* **129,** 307–324.

Besse, J. & Courtillot, V., 2002. Apparent and true polar wander and the geometry of the geomagnetic field over the last 200 Myr, *J. geophys. Res.,* **107,** 2300, doi:10.1029/2000JB000/050.

Biggin, A.J., van Hinsbergen, D.J.J., Langereis, C.G., Straathof, G.B. & Deenen, M.H.L., 2008. Geomagnetic secular variation in the Cretaceous Normal Superchron and in the Jurassic, *Phys. Earth planet. Int.,* **169,** 3–19, doi:10.1016/j.pepi.2008.07.004.

Bohaty, S.M. & Zachos, J.C., 2003. Significant Southern Ocean warming event in the late middle Eocene, *Geology,* **31,** 1017–1020.

Butler, R.F., 1992. *Paleomagnetism: Magnetic Domains to Geologic Terranes,* pp. 238, Blackwell Scientific, Malden, Massachusetts.

Chauvin, A., Perroud, H. & Bazhenov, M.L., 1996. Anomalous low paleomagnetic inclinations from Oligocene-lower Miocene of south-west Tien Shan, central Asia, *Geophys. J. Int.,* **126,** 303–313.

Chen, J.S., Huang, B.C. & Sun, L.S., 2010. New constraints to the onset of the India-Asia collision: paleomagnetic reconnaissance of the Linzizong Group in the Lhasa Block, China, *Tectonophysics,* **489,** 189–209, doi:10.1016/j.tecto.2010.04.024.

Chen, Y., Courtillot, V., Cogné, J.P., Besse, J., Yang, Z.Y. & Enkin, R., 1993. The configuration of Asia prior to the collision of India: Cretaceous paleomagnetic constraints, *J. geophys. Res.,* **98,** 21927–21941.

Chenet, A.L., Courtillot, V., Fluteau, F., Gerard, M., Quidelleur, X., Khadri, S.F.R., Subbarao, K.V. & Thordarson, T., 2009. Determination of rapid Deccan eruptions across the Cretaceous-Tertiary boundary using paleomagnetic secular variation: 2. Constraints from analysis of eight new sections and synthesis for a 3500-m-thick composite section, *J. geophys. Res.,* **114,** doi:10.1029/2008jb005644.

Chung, S.L. *et al*., 2005. Tibetan tectonic evolution inferred from spatial and temporal variations in post-collisional magmatism, *Earth Sci. Rev.,* **68,** 173–196, doi:10.1016/j.earscirev.2004.05.001.

Cogné, J.P., 2003. PaleoMac: a Macintosh application for treating paleomagnetic data and making plate reconstructions, *Geochem. Geophys. Geosyst.* **4,** doi:10.1029/2001GC000227.

Cogné, J.P., Halim, N., Chen, Y. & Courtillot, V., 1999. Resolving the problem of shallow magnetization of Tertiary age in Asia: insights from paleomagnetic data from the Qiangtang, Kunlun, and Qaidam blocks (Tibet, China), and a new hypothesis, *J. geophys. Res.,* **104,** 17715–17734.

Copley, A., Avouac, J.P. & Royer, J.Y., 2010. The India-Asia collision and the Cenozoic slowdown of the Indian plate: implications for the forces driving plate motions, *J. geophys. Res.,* **115,** B03410, doi:10.1029/2009JB006634.

Coward, M.P., Kidd, W.S.F., Yun, P., Shackleton, R.M. & Hu, Z. 1988. The structure of the 1985 Tibet Geotraverse, Lhasa to Golmud, *Phil. Trans. R. Soc. Lond.,* **327,** 307–336.

Cox, A., 1962. Analysis of present geomagnetic field for comparison with palaeomagnetic results, *J. Geomagn. Geoelectr.* **13,** 101–112.

Cox, A., 1970. Latitudinal dependence of the angular dispersion of the geomagnetic field, *Geophys. J. R. astr. Soc.,* **20,** 253–269.

Creer, K.M., 1962. The dispersion of the geomagnetic field due to secular variation and its determination for remote times from palaeomagnetic data, *J. geophys. Res.,* **67,** 3461–3476.

Cunningham, D., 2005. Active intracontinental transpressional mountain building in the Mongolian Altai: defining a new class of orogen, *Earth planet. Sci. Lett.,* **240,** 436–444, doi:10.1016/j.epsl.2005.09.013.

Cunningham, W.D., Windley, B.F., Dorjnamjaa, D., Badamgarov, G. & Saandar, M., 1996. A structural transect across the Mongolian Western Altai: active transpressional mountain building in central Asia, *Tectonics,* **15,** 142–156.

Darby, B.J., Ritts, B.D., Yue, Y.J. & Meng, Q.R., 2005. Did the Altyn Tagh fault extend beyond the Tibetan Plateau?, *Earth planet. Sci. Lett.,* **240,** 425–435, doi:10.1016/j.epsl.2005.09.011.

DeCelles, P.G., Kapp, P., Ding, L. & Gehrels, G.E., 2007a. Late Cretaceous to middle Tertiary basin evolution in the central Tibetan Plateau: changing environments in response to tectonic partitioning, aridification, and regional elevation gain, *Bull. Geol. Soc. Am.,* **119,** 654–680, doi:10.1130/B26074.1.

DeCelles, P.G., Quade, J., Kapp, P., Fan, M.J., Dettman, D.L. & Ding, L., 2007b. High and dry in central Tibet during the Late Oligocene, *Earth planet. Sci. Lett.,* **253,** 389–401, doi:10.1016/j.epsl.2006.11.001.

DeCelles, P.G., Robinson, D.M. & Zandt, G., 2002. Implications of shortening in the Himalayan fold-thrust belt for uplift of the Tibetan Plateau, *Tectonics,* **21,** doi:10.1029/2001TC001322.

Demarest, H.H.J., 1983. Error analysis for the determination of tectonic rotation from paleomagnetic data, *J. geophys. Res.,* **88,** 4321–4328.

Ding, L., Kapp, P. & Wan, X.Q., 2005. Paleocene-Eocene record of ophiolite obduction and initial India-Asia collision, south central Tibet, *Tectonics,* **24,** TC3001, doi:10.1029/3004TC001729.

Ding, L., Kapp, P., Yue, Y.H. & Lai, Q.Z., 2007. Postcollisional calc-alkaline lavas and xenoliths from the southern Qiangtang terrane, central Tibet, *Earth planet. Sci. Lett.,* **254,** 28–38, doi:10.1016/j.epsl.2006.11.019.

Doubrovine, P.V. & Tarduno, J.A., 2008. Linking the Late Cretaceous to Paleogene Pacific plate and the Atlantic bordering continents using plate circuits and paleomagnetic data, *J. geophys. Res.,* **113,** B07104, doi:101029/2008JB005564.

Dunlop, D.J., 2002. Theory and application of the Day plot (M_{rs}/M_s versus H_{cr}/H_c) 1. Theoretical curves and tests using titanomagnetite data, *J. geophys. Res.,* **107,** doi:10.1029/2001JB000486.

Dupont-Nivet, G., Butler, R.F., Yin, A. & Chen, X.H., 2002a. Paleomagnetism indicates no Neogene rotation of the Qaidam Basin in northern Tibet during Indo-Asian collision, *Geology,* **30,** 263–266.

Dupont-Nivet, G., Guo, Z., Butler, R.F. & Jia, C., 2002b. Discordant paleomagnetic direction in Miocene rocks from the central Tarim Basin: evidence for local deformation and inclination shallowing, *Earth planet. Sci. Lett.,* **199,** 473–482.

Dupont-Nivet, G., Hoorn, C. & Konert, M., 2008. Tibetan uplift prior to the Eocene-Oligocene climate transition: evidence from pollen analysis of the Xining Basin, *Geology,* **36,** 987–990, doi:10.1130/G25063A.1.

Dupont-Nivet, G., Krijgsman, W., Langereis, C., Abels, H.A., Dai, S. & Fang, X.M., 2007. Tibetan plateau aridification linked to global cooling at the Eocene-Oligocene transition, *Nature,* **445,** 635–638, doi:10.1038/nature05516.

Dupont-Nivet, G., Lippert, P.C., van Hinsbergen, D.J.J., Meijers, M.J.M. & Kapp, P., 2010a. Palaeolatitude and age of the Indo-Asia collision: palaeomagnetic constraints, *Geophys. J. Int.,* **182,** 1189–1198, doi:10.1111/j.1365-246X.2010.04697.x.

Dupont-Nivet, G., van Hinsbergen, D.J.J. & Torsvik, T.H., 2010b. Persistently low Asian paleolatitudes: implications for the Indo-Asian collision history, *Tectonics,* doi:10.1029/2008TC002437.

Ehlers, T.A. & Poulsen, C.J., 2009. Influence of Andean uplift on climate and paleoaltimetry estimates, *Earth planet. Sci. Lett.,* **281,** 238–248, doi:10.1016/j.epsl.2009.02.026.

England, P.C. & Searle, M.P., 1986. The Cretaceous-Tertiary deformation of the Lhasa block and its implications for crustal thickening in Tibet, *Tectonics,* **5,** 1–14.

Enkin, R., Yang, Z.Y., Chen, Y. & Courtillot, V., 1992. Paleomagnetic constraints on the geodynamic history of the major blocks of China from the Permian to the present, *J. geophys. Res.,* **97,** 13 953–13 989.

Fielding, E., Isacks, B., Barazangi, M. & Duncan, C., 1994. How flat is Tibet? *Geology,* **22,** 163–167.

Fisher, R.A., 1953. Dispersion on a sphere, *Proc. R. Soc. Lond. Math. Phys. Sci.,* **217,** 295–305.

France-Lanord, C. & Derry, L.A., 1997. Organic carbon burial forcing of the carbon cycle from Himalayan erosion, *Nature,* **390,** 65–67.

Galewsky, J., 2009. Rain shadow development during the growth of mountain ranges: an atmospheric dynamics perspective, *J. geophys. Res.,* **114,** F01018, doi:10.1029/2008JF001085.

Ganerød, M., Smethurst, M.A., Rousse, S., Torsvik, T.H. & Prestvik, T., 2008. Reassembling the Paleogene–Eocene North Atlantic igneous province: new paleomagnetic constraints from the Isle of Mull, Scotland, *Earth planet. Sci. Lett.,* **272,** 464–475, doi:10.1016/j.epsl.2008.05.016.

Gilder, S., Chen, Y. & Sen, S., 2001. Oligo-Miocene magnetostratigraphy and environmental magnetism of the Xishuigou section, Subei (Gansu Province, western China): further implication on the shallow inclination of central Asia, *J. geophys. Res.,* **106,** 30 505–30 521.

Gilder, S.A., Gomez, J., Chen, Y. & Cogné, J.P., 2008. A new paleogeographic configuration of the Eurasian landmass resolves a

paleomagnetic paradox of the Tarim Basin (China), *Tectonics,* **27,** TC1012, doi:10.1029/2007TC002155.

Halim, N. *et al*., 1998. New Cretaceous and Early Tertiary paleomagnetic results from Xining-Lanzhou basin, Kunlun and Qiangtang blocks, China: implications on the geodynamic evolution of Asia, *J. geophys. Res.,* **103,** 21 025–21 045.

Halls, H.C., 1978. The use of converging remagnetization circles in paleomagnetism, *Phys. Earth planet. Int.,* **16,** 1–11.

Hankard, F., Cogné, J.P., Kravchinsky, V., Carporzen, L., Bayasgalan, A. & Lkhagvadorj, P., 2007. New Tertiary paleomagnetic poles from Mongolia and Siberia at 40, 30, 20, and 13 Ma: Clues on the inclination shallowing problem in central Asia, *J. geophys. Res.,* **112,** B02101, doi:10.1029/2006JB004488.

Horner-Johnson, B.C., Gordon, R.G. & Argus, D.F., 2007. Plate kinematic evidence for the existence of a distinct plate between the Nubian and Somalian plates along the Southwest Indian Ridge, *J. geophys. Res.,* **112,** doi:10.1029/2006JB004519.

Horton, B.K., Yin, A., Spurlin, M.S., Zhou, J.Y. & Wang, J.H., 2002. Paleocene-Eocene syncontractional sedimentation in narrow, lacustrine-dominated basins of east-central Tibet, *Bull. Geol. Soc. Am.,* **114,** 771–786, doi:10.1130/0016-7606(2002)114<0771:PESSIN>2.0.CO;2.

Insel, N., Poulsen, C.J. & Ehlers, T.A., 2009. Influence of the Andes mountains on South American moisture transport, convection, and precipitation, *Clim. Dyn.,* doi:10.1007/s00382-009-0637-1.

Jarboe, N.A., Coe, R.S., Renne, P.R., Glen, J.M.G. & Mankinen, E.A., 2008. Quickly erupted volcanic sections of the Steens Basalt, Columbia River Basalt Group: Secular variation, tectonic rotation, and the Steens Mountain reversal, *Geochem. Geophys. Geosyst.,* **9,** Q11010, doi:10.1029/2008gc002067.

Johnson, C.L. *et al*., 2008. Recent investigations of the 0–5 Ma geomagnetic field recorded by lava flows, *Geochem. Geophys. Geosyst.* **9,** doi:10.1029/2007GC001696.

Jolivet, M. *et al*., 2007. Mongolian summits: an uplifted, flat, old but still preserved erosion surface, *Geology,* **35,** 871–874, doi:10.1130/G23758A.1.

Kapp, P. *et al*., 2000. Blueschist-bearing metamorphic core complexes in the Qiangtang block reveal deep crustal structure of northern Tibet, *Geology,* **28,** 19–22.

Kapp, P., DeCelles, P.G., Gehrels, G.E., Heizler, M. & Ding, L., 2007a. Geological records of the Lhasa-Qiangtang and Indo-Asian collisions in the Nima area of central Tibet, *Bull. Geol. Soc. Am.,* **119,** 917–932, doi:10.1130/B26033.1.

Kapp, P., DeCelles, P.G., Leier, A.L., Fabijanic, J.M., He, S., Pullen, A., Gehrels, G.E. & Ding, L., 2007b. The Gangdese retroarc thrust belt revealed, *GSA Today,* **17,** 4–9, doi:10.1130/GSAT01707A.1.

Kapp, P., Yin, A., Harrison, T.M. & Ding, L., 2005. Cretaceous-Tertiary shortening, basin development, and volcanism in central Tibet, *Bull. Geol. Soc. Am.,* **117,** 865–878, doi:10.1130/B25595.1.

Kent, D.V. & Muttoni, G., 2008. Equatorial convergence of India and early Cenozoic climate trends, *Proc. Natl. Acad. Sci. U.S.A.,* **105,** 16065–16070, doi:10.1073/pnas.0805382105.

King, R.F., 1955. The remanent magnetism of artificially deposited sediments, *Mon. Not. R. astron. Soc. geophys. Suppl.,* **7,** 115–134.

Kirschvink, J.L., 1980. The least-square line and plane and the analysis of paleomagnetic data, *Geophys. J. R. Astron. Soc.,* **62,** 699–718.

Kono, M., 1997. Distributions of paleomagnetic directions and poles, *Phys. Earth planet. Int.,* **103,** 313–327.

Kuiper, K.F., Deino, A., Hilgen, F.J., Krijgsman, W., Renne, P.R. & Wijbrans, J.R., 2008. Synchronizing Rock Clocks of Earth History, *Science,* **320,** 500–504, doi:10.1126/science.1154339.

Leier, A.L., DeCelles, P.G., Kapp, P. & Ding, L., 2007. The Takena Formation of the Lhasa terrane, southern Tibet: the record of a Late Cretaceous retroarc foreland basin, *Bull. Geol. Soc. Am.,* **119,** 31–48, doi:10.1130/B25974.1.

Liebke, U., Appel, E., Ding, L., Neumann, U., Antolin, B. & Xu, Q.A., 2010. Position of the Lhasa terrane prior to India-Asia collision derived from palaeomagnetic inclinations of 53 Ma old dykes of the Linzhou Basin: constraints on the age of collision and post-collisional shortening within the Tibetan Plateau, *Geophys. J. Int.,* **182,** 1199–1215, doi:10.1111/j.1365-246X.2010.04698.x.

Lin, J. & Watts, D.R. 1988. Paleomagnetic results from the Tibetan Plateau, *Phil. Trans. R. Soc. Lond.,* **327,** 239–262.

Lippert, P.C., van Hinsbergen, D. J. J., Dupont-Nivet, G. & Kapp, P., 2010. Consensus on the Eocene latitude of Lhasa and the age of the Tethyan Himalaya-Asia collision? *EOS, Trans. Am. geophys. Un.* Fall Meeting, Suppl. 91, Abstract T33F-03.

Liu, Z., Zhao, X.X., Wang, C., Liu, S. & Yi, H., 2003. Magnetostratigraphy of Tertiary sediments from the Hoh Xil Basin: implications for the Cenozoic tectonic history of the Tibetan Plateau, *Geophys. J. Int.,* **154,** 233–252, doi:10.1046/j.1365-246X.2003.01986.x.

Lo, C.H., Chung, S.L., Lee, T.Y. & Wu, G.Y., 2002. Age of the Emeishan flood magmatism and relations to Permian-Triassic boundary events, *Earth planet. Sci. Lett.,* **198,** 449–458.

Luterbacher, H.P. *et al*., 2004. The Paleogene Period, in *A Geologic Time Scale 2004,* pp. 384–408, eds Gradstein, F.M. *et al*., Cambridge University Press, Cambridge.

McElhinny, M.W., 1973. Paleomagnetic results from Eurasia, in *Implications of Continental Drift to the Earth Sciences: Proceedings of the NATO Advanced Study Institute,* pp. 77–85, eds Tarling, D.H. & Runcorn, S.K. Academic Press, London.

McElhinny, M.W., 2004. Geocentric axial dipole hypothesis: a least squares perspective, in *Timescales of the Paleomagnetic Field,* pp. 1–12, eds Channell, J.E.T. *et al*., AGU, Washington, DC.

McElhinny, M.W. & McFadden, P.L., 1997. Palaeosecular variation over the past 5 Myr based on a new generalized database, *Geophys. J. Int.,* **131,** 240–252.

McFadden, P.L. & Lowes, F.J., 1981. The discrimination of mean directions drawn from Fisher distributions, *Geophys. J. R. astr. Soc.,* **67,** 19–33.

McFadden, P.L. & McElhinny, M.W., 1988. The combined analysis of remagnetization circles and direct observations in paleomagnetism, *Earth planet. Sci. Lett.,* **87,** 161–172.

McFadden, P.L. & Reid, A.B., 1982. Analysis of palaeomagnetic inclination data, *Geophys. J. R. astr. Soc.,* **69,** 307–319.

Merrill, R.T., McElhinny, M.W. & McFadden, P.L., 1996. *The Magnetic Field of the Earth, Paleomagnetism, the Core, and the Deep Mantle,* p. 531, Academic, New York.

Morris, E.R., Schillinger, W., Coe, R.S., Pluhar, C.J. & Jarboe, N.A., 2009. Automating the 2G superconducting rock magnetometer for single-solenoid alternating field demagnetization, *Geochem. Geophys. Geosyst.,* **10,** Q05Y05, doi:10.1029/2008GC002289.

Murphy, M.A. *et al*., 1997. Did the Indo-Asian collision alone create the Tibetan plateau?, *Geology,* **25,** 719–722, doi:10.1130/0091-7613(1997)025<0719:DTIACA>2.3.CO;2.

Najman, Y. *et al*., 2010. The timing of India-Asia collision: geological, biostratigraphic and paleomagnetic constraints, *J. geophys. Res.,* in press, doi:10.1029/2010JB007673.

Odin, G.S. *et al*., 1982. Interlaboratory standards for dating purposes, in *Numerical Dating in Stratigraphy,* pp. 123–149, ed. Odin, G.S., Wiley, Chichester.

Patriat, P. & Achache, J., 1984. Indian-Asia collision chronology has implications for crustal shortening and driving mechanisms of plates, *Nature,* **311,** 615–621.

Quidelleur, X., Valet, J.P., Courtillot, V. & Hulot, G. 1994. Long-term geometry of the geomagnetic field for the last five million years: an updated secular variation database, *Geophys. Res. Lett.,* **21,** 1639–1642.

Ramstein, G., Fluteau, F., Besse, J. & Joussaume, S., 1997. Effect of orogeny, plate motion and land-sea distribution on Eurasian climate change over the past 30 million years, *Nature,* **386,** 788–795.

Ratschbacher, L., Frisch, W., Chen, C. & Pan, G., 1994. Distributed deformation in southern and western Tibet during and after the India-Asian collision, *J. geophys. Res.,* **99,** 19 971–19 945.

Renne, P.R., Swisher, C.C., Deino, A.L., Karner, D.B., Owens, T.L. & DePaolo, D.J., 1998. Intercalibration of standards, absolute ages and uncertainties in $^{40}Ar/^{39}Ar$ dating, *Chem. Geol.,* **145,** 117–152.

Replumaz, A., Karason, H., van der Hilst, R.D., Besse, J. & Tapponnier, P., 2004. 4-D evolution of SE Asia's mantle from geolog-

ical reconstructions and seismic tomography, *Earth planet. Sci. Lett.,* **221,** 103–115, doi:10.1016/S0012-821X(04)00070-6.

Richter, F.M., Rowley, D.B. & DePaolo, D.J., 1992. Sr isotope evolution of seawater: the role of tectonics, *Earth planet. Sci. Lett.,* **109,** 11–23.

Riisager, J., Riisager, P. & Pedersen, A.K., 2003. Paleomagnetism of large igneous provinces: case-study from West Greenland, North Atlantic igneous province, *Earth planet. Sci. Lett.,* **214,** 409–425, doi:10.1016/S0012-821X(03)00367-4.

Riisager, P., Riisager, J., Abrahamsen, N. & Wagstein, R., 2002. New paleomagnetic pole and magnetostratigraphy of Faroe Islands flood volcanics, North Atlantic igneous province, *Earth planet. Sci. Lett.,* **201,** 261–276.

Rowley, D.B., 1996. Age of initiation of collision between India and Asia: a review of stratigraphic data, *Earth planet. Sci. Lett.,* **145,** 1–13.

Rowley, D.B. & Garzione, C.N., 2007. Stable isotope-based paleoaltimetry, *Ann. Rev. Earth planet. Sci.,* **35,** 463–508, doi:10.1146/annurev.earth.35.031306.140155.

Royer, J.Y., Gordon, R.G. & Horner-Johnson, B.C., 2006. Motion of Nubia relative to Antarctica since 11 Ma: implications for Nubia-Somolia, Pacific-North America, and India-Eurasia motion, *Geology,* **34,** 501–504, doi:10.1130/G22463.1.

Si, J. & Van der Voo, R., 2001. Too-low magnetic inclinations in central Asia: an indication of long-term Tertiary non-dipole field?, *Terra Nova,* **13,** 471–478.

Sobel, E.R., Hilley, G.E. & Strecker, M.R., 2003. Formation of internally-drained contractional basins by aridity-limited bedrock incision, *J. geophys. Res.,* **108,** 2344, doi:10.1029/2002JB001883.

Steiger, R.H. & Jager, E., 1977. Subcommission on geochronology: convention on the use of decay constants in geo- and cosmochronology, *Earth planet. Sci. Lett.,* **36,** 359–362.

Sun, Z.M., Wan, J., Li, H.B., Pei, J.L. & Zhu, Z.M., 2010. New paleomagnetic results of Paleocene volcanic rocks from the Lhasa block: tectonic implications for the collision of India and Asia, *Tectonophysics,* **490,** 257–266, doi:10.1016/j.tecto.2010.05.011.

Tan, X.D., Gilder, S., Kodama, K.P., Jiang, W., Han, Y. L., Zhang, H., Xu, H.H. & Zhou, D., 2010. New paleomagnetic results from the Lhasa block: revised estimation of latitudinal shortening across Tibet and implications for dating the India-Asia collision, *Earth planet. Sci. Lett.,* **293,** 396–404, doi:10.1016/j.epsl.2010.03.013.

Tan, X.D., Kodama, K.P., Chen, H.L., Fang, D.J., Sun, D.J. & Li, Y.G., 2003. Paleomagnetism and magnetic anisotropy of Cretaceous red beds from the Tarim basin, northwest China: evidence for a rock magnetic cause of anomalously shallow paleomagnetic inclinations from central Asia, *J. geophys. Res.,* **108,** 2107, doi:10.1029/2001JB001608.

Tapponnier, P., Peltzer, G., Le Dain, Y., Armijo, R. & Cobbold, P.R., 1982. Propagating extrusion tectonics in Asia: new insights from simple experiments with plasticine, *Geology,* **10,** 611–616.

Tauxe, L., 2005. Inclination flattening and the geocentric axial dipole hypothesis, *Earth planet. Sci. Lett.,* **233,** 247–261, doi:10.1016/j.epsl.2005.01.027.

Tauxe, L. & Kent, D.V., 2004. A simplified statistical model for the geomagnetic field and the detection of shallow bias in paleomagnetic inclinations: was the ancient field dipolar?, in *Timescales of the Paleomagnetic Field,* pp. 101–115, eds Channell, J.E.T. *et al*., American Geophysical Union, Washington, DC.

Tauxe, L., Constable, C., Stokking, L. & Badgley, C., 1990. Use of anisotropy to determine the origin of characteristic remanence in the Siwalik red beds of northern Pakistan, *J. geophys. Res.,* **95,** 4391–4404.

Tauxe, L., Staudigel, H. & Wijbrans, J.R., 2000. Paleomagnetism and $^{40}Ar^{39}Ar$ ages from La Palma in the Canary Islands, *Geochem. Geophys. Geosyst.,* **1,** 1038, doi:10.1029/2000GC000063.

Taylor, M., Yin, A., Ryerson, F.J., Kapp, P. & Ding, L., 2003. Conjugate strike-slip faulting along the Bangong-Nujiang suture zone accommodates coeval east-west extension and north-south shortening in the interior of the Tibetan Plateau, *Tectonics,* **22,** doi:10.1029/2002TC001361.

Torsvik, T.H., Muller, R.D., Van der Voo, R., Steinberger, B. & Gaina, C., 2008. Global plate motion frames: toward a unified model, *Rev. Geophys.,* **46,** RG3004, doi:10.1029/2007RG000227.

Turner, S. *et al*., 1996. Post-collision, shoshonitic volcanism on the Tibetan Plateau: implications for convective thinning of the lithosphere and the source of ocean island basalts, *J. Petrol.,* **37,** 45–71.

Van der Voo, R. & Torsvik, T.H., 2001. Evidence for late Paleozoic and Mesozoic non-dipole fields provides an explanation for the Pangea reconstruction problems, *Earth planet. Sci. Lett.,* **187,** 71–81.

van Hinsbergen, D. J. J., Kapp, P., Dupont-Nivet, G., DeCelles, P.G., Lippert, P.C. & Torsvik, T. H. (submitted), Cenozoic deformation of Tibet, Pamir-Hindu Kush, Tien Shan and Mongolia reviewed and restored: retrodeforming intra-Asian shortening and extrusion north of India, *Earth Sci. Rev.*

Verosub, K.L., 1977. Depositional and postdepositional processes in the magnetization of sediments, *Rev. Geophys.,* **15,** 129–143.

Wang, C.S. *et al*., 2008. Constraints on the early uplift history of the Tibetan plateau, *Proc. Natl. Acad. Sci. U.S.A.,* **105,** 4987–4992, doi:10.1073/pnas.0703595105.

Westphal, M., 1993. Did a large departure from the geocentric axial dipole hypothesis occur during the Eocene? Evidence from the magnetic polar wander path of Eurasia, *Earth planet. Sci. Lett.,* **117,** 15–28.

Yan, M.D., Van der Voo, R., Tauxe, L., Fang, X.M. & Pares, J.M., 2005. Shallow bias in Neogene palaeomagnetic directions from the Guide Basin, NE Tibet, caused by inclination error, *Geophys. J. Int.,* **163,** 944–948, doi:10.1111/j.1365-246X.2005.02802.x.

Yin, A., Dang, Y.Q., Zhang, M., McRivette, M.W., Burgess, W.P. & Chen, X.H., 2007. Cenozoic tectonic evolution of Qaidam basin and its surrounding regions (part 2): Wedge tectonics in the southern Qaidam basin and the Eastern Kunlun Range, *Special Paper 433: Whence the Mountains? Inquiries into the Evolution of Orogenic Systems: A Volume in Honor of Raymond A. Price*, **433,** 369–390, doi:10.1130/2007.2433(18).

Yin, A. & Harrison, T.M., 2000. Geologic evolution of the Himalayan-Tibetan orogen, *Ann. Rev. Earth planet. Sci.,* **28,** 211–280.

Yue, Y.J., Graham, S.A., Ritts, B.D. & Wooden, J.L., 2005. Detrital zircon provenance evidence for large-scale extrusion along the Altyn Tagh fault, *Tectonophysics,* **406,** 165–178, doi:10.1016/j.tecto.2005.05.023.

Zachos, J., Pagani, M., Sloan, L., Thomas, E. & Billups, K., 2001. Trends, rhythms, and aberrations in global climate 65 Ma to present, *Science,* **292,** 686–693.

Zhou, J.X., Xu, F.Y., Wang, T.C., Cao, A.F. & Yin, C.M., 2006. Cenozoic deformation history of the Qaidam Basin, NW China: results from cross-section restoration and implications for Qinghai-Tibet Plateau tectonics, *Earth planet. Sci. Lett.,* **241,** 195–210, doi:10.1016/j.epsl.2005.11.033.

APPENDIX

Table A1. $^{40}Ar/^{39}Ar$ geochonologic data for Wuluan Wula Hu.

Step (°C)	Cummulative ^{39}Ar	per cent ^{39}Ar released	Atmos (per cent)	$^{36}Ar/^{39}Ar$	$^{37}Ar/^{39}Ar$	$^{38}Ar/^{39}Ar$	$^{39}Ar/^{40}Ar$	$^{36}Ar/^{40}Ar$	Age (Ma)
Sample 06X036b[a]									
600	0.007	0.007	80.34	6.94E−02	1.01E+00	2.81E−02	2.55E+01	3.67E+02	37.8 ± 0.8
700	0.054	0.047	38.48	1.19E−02	7.99E−01	1.50E−02	9.01E+00	7.57E+02	41.7 ± 0.5
750	0.120	0.066	27.45	7.03E−03	6.99E−01	1.38E−02	7.40E+00	1.05E+03	40.4 ± 0.5
800	0.197	0.077	16.48	3.66E−03	6.83E−01	1.33E−02	6.28E+00	1.72E+03	39.4 ± 0.5
850	0.324	0.127	10.00	2.11E−03	6.95E−01	1.30E−02	5.73E+00	2.72E+03	38.8 ± 0.5
900	0.453	0.129	8.33	1.74E−03	6.79E−01	1.30E−02	5.60E+00	3.21E+03	38.6 ± 0.5
950	0.566	0.113	8.15	1.69E−03	6.71E−01	1.31E−02	5.54E+00	3.28E+03	38.2 ± 0.5
1000	0.688	0.122	9.23	1.91E−03	7.06E−01	1.33E−02	5.57E+00	2.92E+03	38 ± 0.5
1050	0.778	0.090	12.14	2.56E−03	8.80E−01	1.36E−02	5.72E+00	2.23E+03	37.8 ± 0.5
1100	0.847	0.069	16.29	3.73E−03	1.77E+00	1.43E−02	5.97E+00	1.60E+03	37.6 ± 0.5
1150	0.891	0.044	26.72	7.14E−03	3.73E+00	1.50E−02	6.87E+00	9.63E+02	38 ± 0.5
1200	0.929	0.038	37.93	1.22E−02	6.67E+00	1.60E−02	8.17E+00	6.72E+02	38.3 ± 0.7
1250	0.965	0.036	46.69	1.72E−02	8.80E+00	1.70E−02	9.52E+00	5.52E+02	38.4 ± 0.8
1300	0.987	0.022	68.69	4.00E−02	9.34E+00	2.14E−02	1.62E+01	4.05E+02	38.5 ± 0.8
1450	1.000	0.013	84.62	9.25E−02	1.03E+01	3.05E−02	3.14E+01	3.40E+02	36.7 ± 0.8
Sample 06X109a[b]									
600	0.004	0.004	95.917	9.70E−01	2.05E+00	1.85E−01	2.99E+02	3.08E+02	90.9 ± 2.1
700	0.021	0.017	90.077	2.36E−01	1.46E+00	5.41E−02	7.72E+01	3.28E+02	57.6 ± 0.8
750	0.057	0.036	85.513	1.31E−01	1.12E+00	3.55E−02	4.53E+01	3.45E+02	49.4 ± 0.7
800	0.094	0.037	81.154	9.07E−02	9.52E−01	2.87E−02	3.30E+01	3.64E+02	46.8 ± 0.6
850	0.161	0.067	77.31	6.61E−02	8.22E−01	2.40E−02	2.52E+01	3.82E+02	43.1 ± 0.6
900	0.234	0.073	71.857	4.91E−02	7.53E−01	2.14E−02	2.01E+01	4.10E+02	42.7 ± 0.5
950	0.307	0.073	67.04	3.81E−02	7.38E−01	1.96E−02	1.67E+01	4.39E+02	41.6 ± 0.5
1000	0.410	0.103	62.67	3.06E−02	8.43E−01	1.85E−02	1.44E+01	4.69E+02	40.4 ± 0.5
1050	0.528	0.118	57.354	2.43E−02	9.74E−01	1.75E−02	1.24E+01	5.11E+02	39.9 ± 0.5
1100	0.674	0.146	56.002	2.32E−02	1.40E+00	1.76E−02	1.21E+01	5.21E+02	40.1 ± 0.5
1150	0.771	0.097	55.582	2.33E−02	2.74E+00	1.76E−02	1.21E+01	5.17E+02	40.4 ± 0.5
1200	0.867	0.096	56.52	2.51E−02	5.84E+00	1.81E−02	1.24E+01	4.93E+02	40.7 ± 0.6
1250	0.932	0.065	59.669	2.93E−02	7.98E+00	1.88E−02	1.35E+01	4.62E+02	41.2 ± 0.7
1300	0.975	0.043	64.909	3.70E−02	9.25E+00	2.01E−02	1.58E+01	4.27E+02	42.0 ± 0.8
1450	1.000	0.025	74.921	5.86E−02	9.96E+00	2.40E−02	2.22E+01	3.78E+02	42.1 ± 0.9

Notes: Sample mass = 500.0 mg. J-value = 0.004231388 ± 0.00005125.
[a]Integrated age = 38.6 ± 0.5 Ma Plateau age = 38.4 ± 0.5 (steps 800–1300°C; per cent 39Ar in plateau: 86.7 per cent). Inverse isochron age = 38.4 ± 0.5 Ma (MSWD = 3.47; Initial $^{40}Ar/^{39}Ar$ = 297 ± 7).
[b]Integrated age = 42.0 ± 0.5 Ma. Plateau age = 40.5 ± 0.5 Ma (steps: 1000–1450°C; per cent ^{39}Ar in plateau: 69.3 per cent).

Table A2. Magnetic susceptibility parameters for palaeogene lavas.

Locality	n/N	Mean L	Max L	Mean F	Max F	Mean T	Mean per cent h	Max per cent h
WWH	38/84	1.02 ± 0.02	1.06	1.05 ± 0.02	1.09	0.15 ± 0.47	1.54 ± 0.76	2.91
EJU	93/100	1.01 ± 0.00	1.01	1.01 ± 0.01	1.02	0.31 ± 0.35	1.65 ± 0.57	3.09
NDC	96/106	1.01 ± 0.01	1.04	1.02 ± 0.01	1.05	0.27 ± 0.43	1.11 ± 0.46	2.00

Notes: Anisotropy of magnetic susceptibility (AMS) parameters for South Qiangtang volcanics. WWH, Wulan Wula Hu; EJU, Ejumaima; NDC, Nading Co. *n,* number of samples measured; N, number of samples demagnetized. L, Lineation parameter, k_1/k_2 where k_1, k_2 and k_3 are the maximum, intermediate and minimum eigenvalue of susceptibility, respectively; F, Foliation parameter, k_2/k_3; T, shape factor, $(2k_2 - k_1 - k_3)/(k_1 - k_3)$; per centh: per cent anisotropy, $100(k_1 - k_3)$ (Tauxe *et al*. 1990).

Table A3a. Characteristic remanent directions for Wulan Wula Hu lavas.

Sample	Strike	Dip	Sun compass correction	MDF (mT)	Direction fit	Demagnetization steps	*N*	MAD	D_g	I_g	Used in mean?
Site WWH-01, 34.536°N, 90.227°E											
06 × 034b	0	0	0	5.6	GC	15–120 mT	9	4.7	154.5	47.9	Yes
06 × 035b	0	0	2	5.5	GC	20–140 mT	8	7.7	341.5	−44.9	Yes
06 × 036b	0	0	1	5.5	GC	20–120 mT	9	4.3	340.8	−44.4	Yes
06 × 037b	0	0	2	5.7	GC	25–100 mT	7	6.2	142.9	34.9	Yes
06 × 038a	0	0	2	6.1	LF	20–60 mT	6	2.1	23.1	21.8	Yes
06 × 039b	0	0	3	6.0	GC	25–120 mT	8	2.3	318.1	−40.1	Yes
06 × 040b	0	0	2	6.6	LF	20–60 mT	6	1.6	29.0	2.9	No
06 × 041b	0	0	0	7.3	LF	20–80 mT	7	1.6	24.4	10.4	Yes
06 × 042b	0	0	3	12.6	LF	20–60 mT	6	1.7	17.4	15.7	Yes
06 × 043a*	0	0	2	–	LF	475–560 °C	4	1.4	21.9	35.0	No
06 × 043b	0	0	2	10.3	GC	20–100 mT	8	5.0	123.9	36.4	Yes
Site WWH-02, 34.536°N, 90.227°E											
06 × 054a	0	0	4	8.9	LF	30–100 mT	6	2.8	6.7	36.1	Yes
06 × 055b*	0	0	2	–	GC	100–560 °C	12	3.2	140.0	35.8	No
06 × 055c	0	0	2	9.3	LF	20–80 mT	7	3.6	27.9	29.9	Yes
06 × 056b	0	0	3	9.8	LF	20–50 mT	5	2.7	16.9	26.8	Yes
06 × 057b	0	0	2	11.6	LF	50–100 mT	4	3.3	315.5	39.6	Yes
06 × 058b	0	0	3	6.5	LF	25–60 mT	5	3.2	353.3	68.0	Yes
06 × 059b	0	0	2	8.8	GC	6–30 mT	7	4.1	127.9	32.3	Yes
06 × 060b	0	0	0	7.5	LF	20–80 mT	7	1.1	346.6	27.6	Yes
Site WWH-03, 34.536°N, 90.227°E											
06 × 044b	0	0	7	7.7	LF	20–60 mT	5	2.8	14.8	27.7	Yes
06 × 045b	0	0	6	6.9	LF	20–60 mT	6	1.3	3.7	30.4	Yes
06 × 046b	0	0	8	7.2	LF	20–50 mT	5	1.3	8.5	31.3	Yes
06 × 047b	0	0	−2	6.5	GC	0–120 mT	16	6.4	239.9	12.9	Yes
06 × 048b	0	0	−2	7.3	GC	0–120 mT	16	3.8	234.9	15.0	Yes
06 × 049b	0	0	−5	6.8	LF	25–60 mT	5	1.3	329.4	34.1	Yes
06 × 050b	0	0	−1	7.4	LF	20–60 mT	6	10.5	257.7	69.3	No
06 × 051b	0	0	−1	7.4	LF	25–60 mT	5	0.6	338.5	45.0	Yes
06 × 052a*	0	0	−1	–	LF	500–560 °C	4	2.4	353.2	40.7	No
06 × 052b	0	0	−1	11.9	LF	25–100 mT	7	1.9	354.9	42.2	Yes
06 × 053b	0	0	−1	13.1	LF	25–80 mT	6	0.6	353.0	47.1	Yes
Site WWH-04, 34.536°N, 90.227°E											
06 × 061b	0	0	−4	10.7	LF	40–80 mT	4	1.6	351.2	47.5	Yes
06 × 062b	0	0	0	5.0	LF	30–60 mT	4	1.0	336.4	46.0	Yes
06 × 063b	0	0	0	6.9	LF	20–80 mT	7	1.0	355.3	42.2	Yes
06 × 064b	0	0	3	9.9	LF	20–50 mT	5	1.1	0.1	38.5	Yes
06 × 065a	0	0	3	10.1	LF	15–50 mT	6	0.7	0.5	45.0	Yes
06 × 079b	0	0	0	10.9	LF	20–50 mT	5	0.6	353.5	49.0	Yes
06 × 080c	0	0	−2	13.4	LF	15–60 mT	7	0.3	357.0	41.5	Yes
06 × 081c	0	0	−2	11.9	LF	15–50 mT	6	1.0	356.0	45.4	Yes
06 × 082b	0	0	−1	11.3	LF	25–80 mT	6	1.2	358.2	45.5	Yes
06 × 083a	0	0	−2	11.2	LF	20–60 mT	6	1.2	355.7	43.8	Yes
06 × 084b	0	0	−2	9.7	LF	15–50 mT	6	0.4	3.2	22.5	Yes
Site WWH-05, 34.536°N, 90.227°E											
06 × 072a*	0	0	1	–	LF	540–590 °C	8	3.7	349.0	46.2	No
06 × 072b	0	0	1	14.1	LF	15–130 mT	10	0.6	353.3	46.5	Yes
06 × 073b	0	0	1	11.6	LF	30–100 mT	6	1.0	341.9	50.8	Yes
06 × 074b	0	0	1	10.4	LF	15–60 mT	7	0.7	343.5	46.1	Yes
06 × 075b	0	0	1	12.3	LF	25–140 mT	9	2.9	340.1	47.0	Yes
06 × 076b	0	0	2	11.6	LF	40–100 mT	5	2.7	335.2	51.7	Yes
06 × 077b	0	0	2	13.6	LF	20–100 mT	8	0.7	349.0	47.7	Yes
06 × 078b	0	0	2	15.0	LF	40–80 mT	4	1.3	0.5	53.2	Yes
Site WWH-06											
06 × 066a*	0	0	−1	–	LF	450–560 °C	6	1.0	0.6	50.9	No
06 × 066b	0	0	−1	15.6	LF	40–80 mT	4	1.3	2.8	47.4	Yes
06 × 067b	0	0	−1	16.9	LF	40–120 mT	6	1.6	358.6	45.3	Yes
06 × 068b	0	0	2	18.1	LF	40–100 mT	5	1.3	3.2	48.1	Yes
06 × 069b	0	0	2	18.5	LF	40–100 mT	5	0.9	3.8	44.9	Yes
06 × 070b	0	0	1	16.1	LF	40–100 mT	5	1.9	3.2	43.2	Yes
06 × 071b	0	0	4	21.7	LF	20–80 mT	7	1.3	11.6	47.4	Yes

Table A3a. (*Continued.*)

Sample	Strike	Dip	Sun compass correction	MDF (mT)	Direction fit	Demagnetization steps	*N*	MAD	D_g	I_g	Used in mean?
Site WWH-07, 34.536°N, 90.227°E											
06 × 085a*	0	0	−2	–	LF	520–575 °C	6	2.4	346.6	52.1	No
06 × 085b	0	0	−2	16.7	LF	25–80 mT	6	1.1	352.5	50.4	Yes
06 × 086a	0	0	−2	17.4	LF	40–100 mT	5	2.4	349.5	44.7	Yes
06 × 087b	0	0	−2	17.8	LF	40–100 mT	5	2.7	358.3	46.8	Yes
06 × 088′b	0	0	−2	17.6	LF	40–140 mT	7	2.3	353.9	41.9	Yes
06 × 088b	0	0	−2	13.7	LF	40–100 mT	5	1.6	1.0	47.9	Yes
06 × 106b	0	0	−4	17.5	LF	40–80 mT	4	1.4	347.4	50.4	Yes
06 × 107b	0	0	−2	18.3	LF	40–100 mT	5	1.3	350.7	45.2	Yes
Site WWH-08, 34.536°N, 90.227°E											
06 × 089b	0	0	−2	22.0	LF	25–100 mT	7	0.8	355.0	35.7	Yes
06 × 090b	0	0	−2	23.2	LF	40–120 mT	6	0.6	357.3	50.6	Yes
06 × 091b	0	0	1	27.6	LF	40–140 mT	7	0.9	5.6	39.2	Yes
06 × 092b	0	0	0	34.4	LF	40–140 mT	7	0.6	359.3	48.0	Yes
06 × 093a*	0	0	0	–	LF	475–585 °C	10	3.1	8.0	54.5	No
06 × 093b	0	0	0	36.7	LF	50–160 mT	6	0.6	358.8	52.7	Yes
06 × 104b	0	0	4	19.5	LF	30–100 mT	6	0.9	356.4	43.1	Yes
06 × 105a	0	0	−3	17.8	LF	20–80 mT	7	0.5	2.5	53.3	Yes
Site WWH-09, 34.536°N, 90.227°E											
06 × 094b	0	0	5	14.1	GC	15–130 mT	10	2.6	53.5	−23.6	Yes
06 × 095a	0	0	0	9.8	GC	0–80 mT	14	3.5	114.9	−22.2	No
06 × 096a	0	0	0	16.7	LF	15–60 mT	7	1.1	359.8	48.0	Yes
06 × 098a	0	0	2	19.9	LF	20–100 mT	8	0.9	347.9	48.1	Yes
06 × 098b*	0	0	2	–	LF	400–540 °C	7	2.4	349.0	47.0	No
06 × 099b	0	0	−3	17.6	LF	20–60 mT	6	1.7	354.8	43.6	Yes
06 × 100b	0	0	−2	14.9	LF	20–120 mT	9	1.4	357.6	48.0	Yes
06 × 101b	0	0	2	21.1	LF	20–60 mT	6	0.7	359.9	42.7	Yes
06×102a	0	0	−6	20.0	LF	20–60 mT	6	1.4	15.3	49.7	Yes
Site WWH-10, 34.536°N, 90.227°E											
06 × 108a	0	0	0	19.6	LF	20–80 mT	7	1.2	2.6	46.5	Yes
06 × 109a	0	0	−1	18.7	LF	25–80 mT	6	1.1	355.9	45.2	Yes
06 × 110b	0	0	7	18.5	LF	25–100 mT	7	0.9	6.7	40.2	Yes
06 × 111b	0	0	9	19.6	LF	30–100 mT	5	1.0	5.1	32.2	Yes
06 × 112a*	0	0	3	–	LF	400–520 °C	6	1.4	0.3	35.7	No
06 × 112b	0	0	3	16.8	LF	20–60 mT	6	0.8	359.4	39.9	Yes
06 × 113b	0	0	1	18.0	LF	20–60 mT	6	0.7	354.0	43.9	Yes
06 × 114b	0	0	1	18.5	LF	20–80 mT	6	0.8	352.8	45.9	Yes
06 × 115b	0	0	−2	18.2	LF	20–60 mT	6	0.5	352.4	42.3	Yes

*Notes:** denotes sister sample demagnetized using thermal treatment. MDF, median destructive field of samples demagnetized using alternating field; GC, great-circle fit; LF, vector line fit; Demagnetization Steps, alternating field or temperature step range over which the ChRM is defined. *N*, number of consecutive demagnetization steps used to define the ChRM; MAD, mean angular deviation of the ChRM fit; Dg, Ig, the declination and inclination of the ChRM in geographic coordinates.

Table A3b. Characteristic remanent directions for Ejumaima Lavas.

Sample	Strike	Dip	Sun compass correction	MDF (mT)	Direction fit	Demagnetization steps	*N*	MAD	D_g	I_g	D_s	I_s	Used in mean?
Site EJU-01, 32.219°N, 86.630°E													
07EJ001a	117	21	−4	56.3	LF	60–180 mT	7	0.6	125.7	−40.8	107.3	−40.6	Yes
07EJ001b*	117	21	−4	–	LF	500–600 °C	6	0.5	124.5	−42.1	105.5	−41.3	No
07EJ002a	117	21	0	38.7	LF	60–180 mT	7	0.9	132.7	−39.6	114.4	−42.1	Yes
07EJ003a	117	21	0	41.9	LF	60–160 mT	6	0.8	139.9	−49.2	113.8	−52.9	Yes
07EJ004a	117	21	3	46.8	LF	60–150 mT	5	0.5	144.5	−48.0	119.1	−53.6	Yes
07EJ005a	117	21	2	47.3	LF	80–150 mT	4	1.0	139.0	−36.2	122.4	−41.3	Yes
07EJ006a	117	21	−2	50.0	LF	80–150 mT	4	0.4	135.0	−40.7	115.8	−43.8	Yes
07EJ007a	117	21	0	55.4	LF	60–150 mT	5	0.5	133.1	−38.6	115.4	−41.3	Yes
07EJ008a	117	21	0	49.5	LF	60–150 mT	5	0.4	138.7	−42.3	118.2	−46.6	Yes
Site EJU-02, 32.219°N, 86.630°E													
07EJ009a	117	21	−1	48.9	LF	60–160 mT	6	0.7	134.6	−42.8	114.0	−45.5	Yes
07EJ009b*	117	21	−1	–	LF	500–600 °C	6	0.6	133.1	−45.9	110.4	−47.7	No
07EJ010a	117	21	2	40.8	LF	50–125 mT	5	1.3	150.7	−50.7	122.5	−58.0	Yes
07EJ011a	117	21	1	40.7	LF	80–150 mT	4	2.0	142.8	−53.2	112.4	−57.2	Yes
07EJ012a	117	21	2	43.6	LF	60–150 mT	5	0.5	153.4	−51.2	124.7	−59.4	Yes
07EJ013a	117	21	1	46.6	LF	80–150 mT	4	0.9	131.5	−38.2	114.1	−40.4	Yes
07EJ014a	117	21	2	46.1	LF	80–150 mT	4	0.8	134.5	−49.8	108.5	−51.5	Yes
07EJ015a	117	21	3	48.0	LF	80–150 mT	4	0.6	137.0	−47.1	113.0	−50.1	Yes
07EJ016a	117	21	4	44.8	LF	60–150 mT	5	0.5	138.4	−48.5	113.0	−51.8	Yes
07EJ016b*	117	21	4	–	LF	500–600 °C	6	0.7	140.1	−49.5	113.7	−53.2	Yes
Site EJU-03, 32.219°N, 86.630°E													
07EJ017a	117	21	0	61.0	LF	80–150 mT	4	0.6	136.8	−40.5	117.6	−44.3	Yes
07EJ018a	117	21	0	65.7	LF	80–150 mT	4	0.2	143.2	−50.1	116.0	−54.9	Yes
07EJ018b*	117	21	0	–	LF	520–600 °C	5	0.6	138.5	−47.5	114.0	−51.0	No
07EJ019a	117	21	0	68.6	LF	60–150 mT	5	0.3	130.8	−37.1	114.2	−39.1	Yes
07EJ020a	117	21	3	63.3	LF	60–150 mT	5	0.4	126.8	−54.7	97.6	−52.9	Yes
07EJ021a	117	21	4	48.9	LF	60–125 mT	4	0.5	144.3	−47.9	119.0	−53.4	Yes
07EJ022a	117	21	3	48.0	LF	60–150 mT	5	0.4	155.4	−49.1	129.0	−58.4	Yes
07EJ023a	117	21	8	43.4	LF	60–150 mT	5	0.7	140.7	−49.5	114.2	−53.5	Yes
07EJ023b*	117	21	0	–	LF	500–600 °C	6	0.6	146.9	−47.1	122.3	−53.7	No
07EJ024a	117	21	0	40.8	LF	60–150 mT	5	0.9	125.2	−46.9	102.8	−45.8	Yes
07EJ025a	117	21	0	31.2	LF	60–150 mT	5	1.0	148.4	−50.1	120.9	−56.7	Yes
Site EJU-04, 32.219°N, 86.630°E													
07EJ026a	117	21	3	70.7	LF	60–150 mT	5	0.5	138.3	−53.3	108.4	−55.7	Yes
07EJ027a	117	21	1	65.4	LF	60–150 mT	5	0.3	126.7	−48.7	102.7	−47.8	Yes
07EJ028a	117	21	5	71.5	LF	60–150 mT	5	0.7	143.7	−56.9	108.7	−60.5	Yes
07EJ029a	117	21	0	73.3	LF	60–150 mT	5	0.7	131.9	−61.0	94.2	−59.5	Yes
07EJ029b*	117	21	0	–	LF	500–600 °C	6	0.5	126.4	−62.8	88.1	−59.0	No
07EJ030a	117	21	0	69.9	LF	60–150 mT	5	0.4	128.4	−56.6	96.9	−54.9	Yes
07EJ031a	117	21	0	68.0	LF	60–150 mT	5	0.6	143.2	−56.3	109.1	−59.8	Yes
Site EJU-05, 32.219°N, 86.630°E													
07EJ032a	117	21	−2	40.4	LF	30–100 mT	5	2.3	151.6	−50.5	123.6	−58.2	Yes
07EJ033a	117	21	−3	33.6	LF	30–80 mT	5	2.3	156.1	−48.5	130.3	−58.1	Yes
07EJ034a	117	21	−4	38.2	LF	30–120 mT	9	1.1	153.9	−48.7	127.8	−57.5	Yes
07EJ034b*	117	21	−4	–	LF	500–600 °C	6	0.9	151.7	−48.4	125.8	−56.5	No
07EJ035a	117	21	−2	47.8	LF	15–100 mT	6	4.9	150.8	−48.7	124.6	−56.4	Yes
07EJ036a	117	21	0	36.1	LF	30–80 mT	5	2.1	160.0	−49.6	133.4	−60.4	Yes
07EJ037a	117	21	0	39.7	LF	30–100 mT	8	1.3	138.6	−51.3	110.6	−54.2	Yes
07EJ038a	117	21	0	42.2	LF	30–90 mT	7	0.7	160.7	−52.6	130.7	−63.1	Yes

Table A3b. (*Continued.*)

Sample	Strike	Dip	Sun compass correction	MDF (mT)	Direction fit	Demagnetization steps	*N*	MAD	D_g	I_g	D_s	I_s	Used in mean?
Site EJU-06, 32.219°N, 86.630°E													
07EJ039a	117	21	0	39.9	LF	50–120 mT	8	2.2	156.3	−49.4	129.6	−58.9	Yes
07EJ040a	117	21	0	43.8	LF	60–110 mT	6	1.0	163.2	−50.3	136.3	−62.1	Yes
07EJ040b*	117	21	0	–	LF	500–600 °C	5	1.1	167.7	−51.7	140.0	−64.8	No
07EJ041a	117	21	0	37.7	LF	30–110 mT	9	3.3	150.3	−47.0	125.7	−54.8	Yes
07EJ042a	117	21	0	37.0	LF	40–100 mT	7	2.4	168.9	−52.9	139.9	−66.2	Yes
07EJ043a	117	21	0	28.6	LF	30–90 mT	7	5.5	160.2	−40.0	141.9	−52.0	Yes
07EJ044a	117	21	0	40.3	LF	30–110 mT	9	1.0	177.9	−50.5	155.1	−66.9	Yes
07EJ045a	117	21	0	40.7	LF	50–100 mT	6	1.5	153.8	−54.3	121.3	−62.1	Yes
Site EJU-07, 32.219°N, 86.630°E													
07EJ046a	117	21	1	45.6	LF	50–110 mT	7	2.3	160.3	−49.6	133.8	−60.5	Yes
07EJ047a	117	21	0	39.4	LF	40–110 mT	8	1.3	150.9	−43.9	129.0	−52.3	Yes
07EJ048a	117	21	0	39.1	LF	40–110 mT	8	1.6	154.0	−46.8	129.7	−55.9	Yes
07EJ049a	117	21	0	46.4	LF	60–110 mT	6	0.9	153.8	−49.1	127.3	−57.8	Yes
07EJ050a	117	21	0	48.2	LF	50–110 mT	7	0.5	160.5	−46.9	136.7	−58.2	Yes
07EJ051a	117	21	0	46.8	LF	50–110 mT	7	0.5	158.2	−48.6	132.5	−58.9	Yes
07EJ052a	117	21	0	29.5	LF	30–100 mT	8	2.3	155.6	−40.8	136.3	−51.2	Yes
Site EJU-08, 32.219°N, 86.630°E													
07EJ053a	117	21	0	7.1	LF	50–140 mT	8	4.8	151.9	−41.4	131.9	−50.5	Yes
07EJ054a	117	21	0	37.3	LF	50–110 mT	5	3.3	153.9	−57.3	117.2	−64.4	Yes
07EJ055a	117	21	0	26.9	LF	30–110 mT	7	2.0	154.2	−48.5	128.3	−57.4	Yes
07EJ056a	117	21	0	22.7	LF	50–95 mT	4	7.8	149.7	−51.1	121.1	−58.0	Yes
07EJ057a	117	21	0	19.3	LF	60–120 mT	4	7.0	151.0	−39.3	132.4	−48.3	Yes
07EJ058a	117	21	0	7.9	GC	25–50 mT	6	10.4	185.7	43.2	190.2	23.3	Yes
Site EJU-09, 32.219°N, 86.630°E													
07EJ060a	117	21	0	31.4	LF	45–120 mT	6	3.5	145.2	−49.2	118.7	−54.8	Yes
07EJ061a	117	21	0	32.0	LF	40–100 mT	6	1.9	146.8	−44.3	124.5	−51.2	Yes
07EJ062a	117	21	0	29.4	LF	40–100 mT	5	2.7	152.6	−35.3	136.6	−45.2	Yes
07EJ063a	117	21	0	34.5	LF	40–100 mT	5	3.2	146.3	−51.7	117.2	−57.3	Yes
07EJ064a	117	21	0	31.3	LF	50–120 mT	5	1.2	156.6	−42.7	135.9	−53.2	Yes
07EJ065a	117	21	0	43.3	LF	40–120 mT	6	0.3	150.3	−52.2	120.5	−59.1	Yes
Site EJU-10, 32.219°N, 86.805°E													
07EJ066a	117	21	0	25.1	LF	20–80 mT	6	2.1	152.5	−51.9	123.0	−59.7	Yes
07EJ066b*	117	21	0	–	LF	520–580 °C	4	2.3	146.1	−50.0	118.8	−55.8	No
07EJ067a	117	21	2	34.5	LF	20–80 mT	6	1.1	192.8	−48.8	180.7	−68.6	No
07EJ069a	117	21	0	26.4	LF	20–80 mT	6	1.3	151.4	−53.3	120.2	−60.4	Yes
07EJ070a	117	21	0	19.7	LF	20–80 mT	6	3.6	143.9	−50.4	116.3	−55.4	Yes
07EJ070b*	117	21	0	–	LF	500–560 °C	4	1.3	152.1	−54.8	119.0	−61.8	No
07EJ071a	117	21	1	23.6	LF	30–80 mT	5	2.6	161.0	−52.8	130.8	−63.4	Yes
07EJ072a	117	21	1	24.2	LF	20–60 mT	5	2.4	149.2	−53.0	118.5	−59.4	Yes
Site EJU-11, 32.219°N, 86.805°E													
07EJ073a	117	21	0	38.7	LF	20–80 mT	6	2.4	152.9	−54.0	120.8	−61.5	Yes
07EJ074a	117	21	0	40.2	LF	20–80 mT	6	3.9	151.8	−52.1	122.0	−59.6	Yes
07EJ075a	117	21	1	39.2	LF	20–80 mT	6	4.2	153.4	−63.6	105.2	−68.6	Yes
07EJ076a	117	21	0	40.0	LF	20–60 mT	5	3.0	155.2	−58.8	116.0	−66.0	Yes
07EJ077a	117	21	0	43.0	LF	20–80 mT	6	4.1	149.1	−57.6	112.4	−62.9	Yes
07EJ078a	117	21	0	42.6	LF	20–100 mT	7	0.6	156.5	−52.8	126.0	−61.8	Yes
07EJ079a	117	21	0	40.6	LF	20–100 mT	7	0.6	144.4	−48.9	118.2	−54.3	Yes

Table A3b. (*Continued.*)

Sample	Strike	Dip	Sun compass correction	MDF (mT)	Direction fit	Demagnetization steps	*N*	MAD	D_g	I_g	D_{s-2}	I_{s-2}	Used in mean?
Site EJU-12, 32.219°N, 86.805°E													
07EJ080a	117	21	0	80.1	LF	50–160 mT	7	0.5	172.6	−45.0	153.1	−60.4	No
07EJ081a	117	21	0	76.5	LF	60–150 mT	6	0.6	157.4	−56.0	122.5	−64.7	Yes
07EJ081b*	117	21	0	–	LF	520–600 °C	5	0.4	163.0	−54.5	130.7	−65.5	No
07EJ083a	117	21	0	69.9	LF	40–125 mT	6	0.3	150.9	−57.6	114.0	−63.6	Yes
07EJ085a	117	21	0	43.9	LF	40–150 mT	7	0.4	157.9	−57.2	121.2	−65.8	Yes
07EJ086a	117	21	0	40.4	LF	40–125 mT	6	0.6	159.2	−57.7	121.6	−66.6	Yes
Site EJU-13, 32.219°N, 86.805°E													
07EJ087a	117	21	0	39.3	LF	50–150 mT	6	0.9	147.3	−62.9	102.1	−66.0	Yes
07EJ088a	117	21	0	41.8	LF	60–150 mT	5	1.4	146.9	−63.0	101.7	−65.9	Yes
07EJ089a	117	21	0	41.9	LF	60–175 mT	6	0.6	140.1	−52.4	110.9	−55.6	Yes
07EJ090a	117	21	0	45.1	LF	60–150 mT	5	0.6	135.9	−53.0	106.6	−54.6	Yes
07EJ090b*	117	21	0	–	LF	500–600 °C	6	0.5	143.6	−52.8	113.5	−57.2	No
07EJ091a	117	21	0	44.5	LF	60–150 mT	5	1.0	144.2	−51.9	115.0	−56.7	Yes
07EJ092a	117	21	0	45.0	LF	60–150 mT	5	0.6	139.8	−54.3	108.6	−57.1	Yes
07EJ093a	117	21	0	41.6	LF	60–150 mT	5	0.5	142.8	−54.1	111.4	−58.0	Yes
Site EJU-14, 32.219°N, 86.805°E													
07EJ094a	117	21	0	43.9	LF	60–150 mT	5	1.4	151.4	−50.4	123.5	−58.0	Yes
07EJ094b*	117	21	−1	–	LF	500–580°C	5	0.7	143.8	−51.6	115.0	−56.3	No
07EJ095a	117	21	−1	44.8	LF	60–150 mT	5	1.4	148.4	−51.8	119.1	−58.1	Yes
07EJ096a	117	21	0	46.5	LF	60–150 mT	5	2.4	154.7	−50.9	126.3	−59.6	Yes
07EJ097a	117	21	0	64.5	LF	60–150 mT	5	1.7	154.3	−51.7	125.0	−60.1	Yes
07EJ097b*	117	21	0	–	LF	500–580 °C	5	1.2	153.6	−52.0	124.0	−60.1	No
07EJ098a	117	21	0	64.5	LF	60–150 mT	5	0.9	159.1	−56.5	123.5	−65.7	Yes
07EJ099a	117	21	0	75.9	LF	60–150 mT	5	0.7	150.0	−58.1	112.4	−63.6	Yes
07EJ100a	117	21	0	67.6	LF	60–150 mT	5	0.7	146.6	−54.3	114.5	−59.5	Yes
07EJ100b*	117	21	0	–	LF	500–585 °C	5	0.7	152.1	−55.5	118.1	−62.4	No

Note: Notation same as in Table A3a. D_s, I_s, declination and inclination of ChRM in stratigraphic coordinates.

Table A3c. Characteristic remanent directions for Nading Co Lavas.

Sample	Strike	Dip	Sun compass correction	MDF (mT)	Direction fit	Demagnetization steps	N	MAD	D_g	I_g	D_s	I_s	Used in mean?
Site NDC-11, 32.685°N, 85.604°E													
07ND100a	148	22	−3	46.5	LF	60–140 mT	5	3.9	26.3	32.2	14.5	49.8	Yes
07ND101a	148	22	−3	38.2	LF	50–100 mT	4	3.1	20.2	29.9	8.2	46.0	Yes
07ND102a	148	22	−3	38.0	LF	80–180 mT	5	2.9	17.7	35.3	2.3	50.3	Yes
07ND104a	148	22	−3	35.8	LF	50–100 mT	4	1.8	13.8	28.9	1.3	43.1	Yes
07ND104b*	148	22	−3	–	LF	450–685 °C	5	0.4	22.4	34.7	8.3	51.1	No
07ND105a	148	22	−4	29.0	LF	40–080 mT	4	1.6	17.2	30.6	4.3	45.7	Yes
07ND106a	148	22	−4	39.1	LF	40–080 mT	4	2.7	10.3	17.4	2.4	31.2	Yes
Site NDC-12, 32.685°N, 85.604°E													
07ND107a	148	22	−1	38.5	LF	50–150 mT	6	3.7	20.3	25.6	10.2	41.9	Yes
07ND107b*	148	22	−1	–	LF	450–685 °C	5	0.6	22.4	28.3	11.5	45.1	No
07ND108a	148	22	0	39.3	LF	50–100 mT	4	2.6	10.1	15.0	3.1	28.9	Yes
07ND109a	148	22	−1	42.3	LF	50–100 mT	4	2.7	38.9	29.9	31.7	50.3	Yes
07ND110a	148	22	−1	42.3	LF	50–100 mT	4	1.5	12.6	25.4	1.6	39.4	Yes
07ND111a	148	22	−2	51.1	LF	50–150 mT	6	1.8	29.9	33.5	18.4	51.9	Yes
07ND112a	148	22	−3	50.7	LF	50–150 mT	6	1.4	18.7	33.8	4.4	49.2	Yes
07ND113a	148	22	−3	50.2	LF	60–150 mT	5	3.3	11.4	37.6	353.6	50.3	Yes
Site NDC-13, 32.685°N, 85.604°E													
07ND114a	148	22	4	56.3	LF	50–120 mT	5	2.8	25.3	31.2	13.7	48.6	Yes
07ND114b*	148	22	4	–	LF	450–685 °C	5	0.9	31.2	32.4	20.6	51.2	No
07ND115a	148	22	4	54.1	LF	50–120 mT	5	2.0	26.5	39.7	10.5	56.9	Yes
07ND116a	148	22	1	58.0	LF	50–120 mT	5	0.9	30.5	44.4	12.6	62.4	Yes
07ND117a	148	22	−1	54.0	LF	50–120 mT	5	1.7	26.4	36.4	12.4	53.8	Yes
07ND118a	148	22	2	55.3	LF	50–100 mT	4	2.0	4.8	32.2	349.8	43.2	No
07ND119a	148	22	2	52.7	LF	50–150 mT	6	3.3	31.9	29.8	22.6	48.8	Yes
07ND120a	148	22	3	55.1	LF	50–100 mT	4	3.1	28.3	47.1	7.1	64.3	Yes
Site NDC-14, 32.685°N, 85.604°E													
07ND121a	148	22	1.6	58.6	LF	40–080 mT	4	3.1	39.3	30.6	32.0	51.0	Yes
07ND122a	148	22	−2.0	57.8	LF	50–100 mT	4	5.1	19.5	19.9	11.5	36.2	No
07ND123a	148	22	0.6	57.0	LF	50–100 mT	4	5.2	25.2	26.6	15.5	44.2	Yes
07ND124a	148	22	0.0	55.3	LF	40–100 mT	5	3.1	34.9	27.7	27.1	47.4	Yes
07ND125a	148	22	1.1	54.7	LF	40–100 mT	4	2.2	31.5	26.0	23.4	45.1	Yes
07ND126a	148	22	2.8	55.8	LF	50–120 mT	5	4.1	48.2	43.3	41.1	64.8	No
07ND127a	148	22	−2.8	54.9	LF	50–100 mT	4	4.2	23.1	25.8	13.4	42.9	Yes
07ND128a	148	22	0.0	58.9	LF	50–120 mT	5	4.9	43.1	40.0	34.2	60.8	No
Site NDC-15, 32.685°N, 85.604°E													
07ND129a	148	22	−1.1	41.9	LF	50–100 mT	4	1.3	34.1	12.5	30.1	32.3	Yes
07ND130a	148	22	1.0	46.2	LF	50–100 mT	4	1.4	30.5	4.6	27.8	23.9	Yes
07ND131a	148	22	−1.9	48.5	LF	50–100 mT	4	0.9	30.0	13.6	25.2	32.6	Yes
07ND132a	148	22	0.0	50.6	LF	50–120 mT	5	2.5	32.1	3.5	29.7	23.1	Yes
07ND133a	148	22	0.0	48.8	LF	50–100 mT	4	1.4	28.7	18.7	22.3	37.4	Yes
07ND134a	148	22	0.0	48.8	LF	50–150 mT	6	1.2	34.5	29.8	25.9	49.4	Yes
07ND135a	148	22	0.0	49.5	LF	50–120 mT	5	1.1	29.9	21.4	22.9	40.2	Yes
07ND136a	148	22	0.0	51.1	LF	50–120 mT	5	1.9	21.6	18.3	14.4	35.3	Yes
07ND136b*	148	22	0.0	–	LF	500–685 °C	4	0.8	16.4	17.4	8.9	33.0	No
Site NDC-16, 32.685°N, 85.604°E													
07ND137a	148	22	0	48.6	LF	50–120 mT	5	1.5	17.3	14.5	10.9	30.5	Yes
07ND138a	148	22	0	48.6	LF	50–120 mT	5	2.3	31.5	24.5	23.9	43.6	Yes
07ND139a	148	22	0	49.1	LF	50–120 mT	5	2.1	25.4	20.9	17.8	38.7	Yes
07ND140a	148	22	2	49.8	LF	50–120 mT	5	0.9	32.5	25.4	24.8	44.7	Yes
07ND141a	148	22	0	47.5	LF	50–120 mT	5	1.9	26.7	22.1	18.9	40.2	Yes
07ND141b*	148	22	0	–	LF	450-T685 °C	5	0.8	23.1	18.4	16.0	35.7	No
07ND142a	148	22	0	49.4	LF	60–110 mT	5	4.0	20.4	19.4	12.6	36.0	Yes
07ND143a	148	22	0	51.9	LF	60–120 mT	6	2.9	23.4	17.3	16.7	34.8	Yes
Site NDC-17, 32.685°N, 85.604°E													
07ND144a	148	22	0	28.2	LF	60–120 mT	6	6.2	29.7	25.8	21.3	44.5	Yes
07ND145a	148	22	−5	21.1	LF	60–150 mT	7	2.2	25.9	32.4	13.9	49.9	Yes
07ND146a	148	22	−3	9.5	LF	60–120 mT	6	1.7	15.6	26.5	4.4	41.4	Yes
07ND147a	148	22	−3	16.6	LF	60–120 mT	6	3.3	25.0	26.4	15.4	43.9	Yes
07ND148a	148	22	−2	10.0	LF	50–120 mT	7	2.4	25.5	29.2	14.8	46.7	Yes

Table A3c. (*Continued.*)

Sample	Strike	Dip	Sun compass correction	MDF (mT)	Direction fit	Demagnetization steps	*N*	MAD	D_g	I_g	D_s	I_s	Used in mean?
07ND149a	148	22	−4	9.5	LF	60–120 mT	6	5.1	90.9	37.3	105.8	54.3	No
07ND150a	148	22	0	9.8	LF	50–110 mT	6	3.9	24.1	23.1	15.6	40.5	Yes
07ND151a	148	22	0	9.2	LF	60–120 mT	6	4.4	28.9	30.3	18.6	48.6	Yes
Site NDC-18, 32.685°N, 85.604°E													
07ND152a	148	22	0	45.9	LF	60–110 mT	5	1.5	2.3	27.0	349.9	37.5	Yes
07ND153a	148	22	0	47.3	LF	60–120 mT	6	1.6	3.5	33.6	347.6	44.0	Yes
07ND155a	148	22	0	49.8	LF	80–120 mT	5	2.3	3.4	30.7	349.1	41.3	Yes
07ND156a	148	22	0	46.1	LF	80–120 mT	5	3.3	2.0	24.0	351.0	34.6	Yes
07ND157a	148	22	−1	52.3	LF	80–120 mT	5	2.5	1.8	28.6	348.6	38.8	Yes
07ND158a	148	22	0	48.0	LF	80–120 mT	5	6.6	7.2	43.0	345.0	53.7	Yes
Site NDC-20, 32.685°N, 85.604°E													
07ND166a	148	22	−2	46.0	LF	50–100 mT	4	2.9	21.0	12.5	15.5	29.5	Yes
07ND167a	148	22	0	43.4	LF	80–120 mT	5	11.0	29.8	15.2	24.6	34.2	Yes
07ND168a	148	22	0	48.8	LF	60–100 mT	4	4.8	27.2	13.1	22.2	31.6	Yes
07ND169a	148	22	0	50.7	LF	60–110 mT	5	6.5	32.1	17.1	26.7	36.5	Yes
07ND170a	148	22	0	49.6	LF	60–120 mT	6	5.9	28.5	17.9	22.3	36.5	Yes
07ND171a	148	22	0	51.7	LF	60–100 mT	4	3.7	29.5	15.0	24.3	33.9	Yes
07ND172a	148	22	0	48.0	LF	50–100 mT	4	2.7	23.7	5.3	20.4	23.2	Yes
Site NDC-21, 32.737°N, 85.575°E													
07ND173a	2	15	0	18.3	LF	40–090 mT	5	5.5	28.8	31.0	35.5	23.4	Yes
07ND174a	2	15	0	9.8	LF	50–150 mT	6	3.6	10.0	31.5	18.5	28.3	Yes
07ND175a	2	15	5	7.2	LF	60–120 mT	6	2.7	10.9	34.9	20.5	31.3	Yes
07ND176a	2	15	0	11.4	LF	60–110 mT	5	12.9	6.6	35.0	16.5	32.5	Yes
07ND177a	2	15	0	11.1	LF	50–120 mT	5	9.3	18.2	33.9	26.8	28.6	Yes
07ND177b*	2	15	0	–	LF	450–560 °C	4	0.9	17.7	36.5	27.2	31.2	No
07ND178a	2	15	0	28.4	LF	80–180 mT	8	6.5	27.0	30.8	33.8	23.6	Yes
07ND179a	2	15	0	28.1	LF	60–110 mT	5	6.7	21.3	36.3	30.4	30.2	Yes
07ND180a	2	15	0	27.0	LF	50–100 mT	5	7.1	16.1	30.0	23.6	25.4	Yes
Site NDC-22, 32.737°N, 85.575°E													
07ND181a	2	15	0	30.9	LF	60–120 mT	6	5.5	48.5	42.4	55.7	30.8	No
07ND182a	2	15	0	4.3	GC	0–120 mT	16	6.4	148.1	44.8	138.2	35.3	Yes
07ND183a	2	15	0	18.2	LF	40–120 mT	8	2.7	41.1	38.4	48.4	28.2	No
07ND184a	2	15	0	27.9	LF	60–110 mT	5	7.7	12.2	39.2	23.2	35.1	Yes
07ND185a	2	15	0	7.1	LF	40–100 mT	6	4.3	28.6	36.8	37.0	29.1	Yes
07ND187a	2	15	0	7.7	LF	40–090 mT	5	7.8	14.2	45.3	27.5	40.4	Yes
07ND188a	2	15	0	7.1	LF	60–100 mT	4	4.5	15.0	31.3	23.0	26.9	Yes
Site NDC-22, 32.737°N, 85.575°E													
07ND189a	2	15	5	8.7	GC	0–120 mT	16	2.4	253.1	33.1	248.5	47.1	Yes
07ND190a	2	15	7	7.3	LF	40–100 mT	6	2.1	17.9	37.6	27.8	32.2	Yes
07ND191a	2	15	6	12.9	GC	0–140 mT	18	7.4	285.2	18.6	287.0	33.2	Yes
07ND193a	2	15	1	31.2	LF	50–090 mT	4	2.1	33.0	29.4	38.9	21.0	Yes
07ND194a	2	15	3	10.0	LF	40–100 mT	6	2.4	25.3	30.0	32.1	23.2	Yes
Site NDC-24, 32.737°N, 85.575°E													
07ND195a	2	15	5	9.4	LF	40–100 mT	6	3.2	20.7	29.0	27.6	23.3	Yes
07ND196a	2	15	6	5.9	LF	40–090 mT	5	5.9	23.6	27.7	29.9	21.4	Yes
07ND197a	2	15	7	6.3	LF	30–090 mT	6	3.8	16.2	31.8	24.3	27.1	Yes
07ND198a	2	15	7	7.3	LF	30–100 mT	6	2.2	20.6	29.7	27.7	24.0	Yes
07ND199a	2	15	6	6.7	LF	50–100 mT	5	2.6	21.4	25.0	27.1	19.3	Yes
07ND200a	2	15	6	7.9	LF	40–080 mT	4	6.0	23.3	27.5	29.5	21.3	Yes
Site NDC-25, 32.737°N, 85.575°E													
07ND201a	2	15	6	24.3	LF	50–100 mT	5	3.7	19.3	18.2	23.3	13.2	Yes
07ND202a	2	15	6	23.9	LF	50–100 mT	5	3.5	17.3	16.7	21.0	12.2	Yes
07ND203a	2	15	6	25.0	LF	50–100 mT	4	3.2	18.1	19.0	22.4	14.3	Yes
07ND204a	2	15	5	24.5	LF	50–100 mT	5	3.7	13.7	23.8	19.6	20.0	Yes
07ND205a	2	15	5	26.5	LF	50–090 mT	4	4.2	15.6	21.5	20.7	17.3	Yes
07ND206a	2	15	0	31.5	LF	50–120 mT	5	3.7	10.8	26.1	17.5	22.9	Yes
07ND206b*	2	15	0	–	LF	450–560 °C	4	2.1	6.1	25.8	13	23.8	No

Note: Characteristic remanent magnetization directions (ChRMs) for Ejumaima Lavas. Notation same as in Tables A3a and A3b.

Table A4. Palaeomagnetic site mean directions from 30–40 Ma lavas in Mongolia.

Site	N	D_g	I_g	k	α_{95}	Reference pole 3	Reference pole 4	Reference
Bogd Plateau (30.3 ± 0.1 Ma) 44.7°N, 102.3°E								
BP-II-IV	14	186.3	−48.0	254.4	2.5		*	Dupont-Nivet *et al.* (2010b)
BP-III-V-VI	18	190.0	−53.4	218.3	2.3		*	Dupont-Nivet *et al*. (2010b)
BP-IX	7	182.5	−45.0	506.0	2.7		*	Dupont-Nivet *et al*. (2010b)
BP-VII	7	188.3	−46.5	197.6	4.3		*	Dupont-Nivet *et al*. (2010b)
BP-X	7	196.9	−59.9	451.9	2.8		*	Dupont-Nivet *et al*. (2010b)
BP-XI	7	189.9	−49.6	122.3	5.5		*	Dupont-Nivet *et al*. (2010b)
BP-XII	7	198.2	−49.9	109.3	5.8		*	Dupont-Nivet *et al*. (2010b)
BP-XIV	6	170.1	−39.6	157.1	4.8		*	Dupont-Nivet *et al*. (2010b)
Jaran Plateau (∼30 Ma) 44.8°N, 100.6°E								
JP-I	7	157.7	−59.5	153.5	4.9		*	Dupont-Nivet *et al.* (2010b)
JP-II	7	175.8	−64.8	519.8	2.7		*	Dupont-Nivet *et al.* (2010b)
JP-III	7	116.1	−75.7	254.7	3.8		*	Dupont-Nivet *et al.* (2010b)
JP-IV	6	142.8	−65.9	51.0	9.5		*	Dupont-Nivet *et al.* (2010b)
JP-V	5	170.3	−74.7	121.5	7.0		*	Dupont-Nivet *et al.* (2010b)
JP-VII	6	186.2	−56.7	132.2	5.8		*	Dupont-Nivet *et al.* (2010b)
Khaton Sudal (39.4. ± 0.6 Ma) 44.5°N, 101.4°E								
KS-87	9	193.1	−64.1	37.1	8.6	*		Hankard *et al.* (2007)
KS-88	9	196.5	−58.2	969.1	1.7	*	*	Hankard *et al*. (2007)
KS-89	9	214.9	−59.2	149.5	4.2	*	*	Hankard *et al*. (2007)
KS-90	9	218.5	−53.3	351.5	2.8	*	*	Hankard *et al.* (2007)
KS-91	6	199.6	−48.8	65.5	8.3	*	*	Hankard *et al*. (2007)
KS-93	8	199.0	−68.0	293.5	3.2	*	*	Hankard *et al*. (2007)
KS-94	7	193.9	−53.0	360.3	3.2	*	*	Hankard *et al*. (2007)
KS-I	5	191.0	−47.0	78.2	8.7		*	Dupont-Nivet *et al*. (2010b)
KS-II	7	169.5	−57.8	551.3	2.6		*	Dupont-Nivet *et al*. (2010b)
KS-III	6	197.5	−60.7	129.2	5.9		*	Dupont-Nivet *et al*. (2010b)
KS-IV	7	195.6	−56.7	200.9	4.3		*	Dupont-Nivet *et al*. (2010b)
KS-V	6	195.2	−31.2	58.8	7.9		*	Dupont-Nivet *et al*. (2010b)
KS-VI	7	186.5	−50.2	940.9	2.0		*	Dupont-Nivet *et al*. (2010b)
KS-VII	6	179.2	−59.3	632.5	2.7		*	Dupont-Nivet *et al*. (2010b)
Taatsyn Gol 1 (31.5 ± 0.7 Ma) 45.4°N, 101.3°E								
TG1–1	6	189.0	−55.0	204.4	4.7	*	*	Hankard *et al*. (2007)
TG1-2	7	180.7	−43.8	139.2	5.4	*	*	Hankard *et al*. (2007)
TG1-20	8	176.2	−57.8	282.4	3.6	*	*	Hankard *et al*. (2007)
TG1-21	6	177.1	−52.4	629.3	2.2	*	*	Hankard *et al*. (2007)
TG1-22	6	179.5	−54.8	378.0	3.5	*	*	Hankard *et al*. (2007)
TG1-23	7	161.9	−58.4	117.0	6.2	*	*	Hankard *et al*. (2007)
TG1-24	4	178.3	−54.3	605.7	2.5	*	*	Hankard *et al*. (2007)
TG1-3	6	191.0	−52.1	111.2	6.4	*	*	Hankard *et al*. (2007)
TG1-75	9	185.8	−46.9	61.7	6.6	*	*	Hankard *et al*. (2007)
TG1-76-78	16	177.5	−53.7	44.5	5.6	*		Hankard *et al*. (2007)
TG1-77-79	9	181.3	−50.4	110.5	3.3	*	*	Hankard *et al*. (2007)
TG1-80	9	181.1	−51.6	27.8	9.9	*		Hankard *et al*. (2007)
TG1-81	10	185.6	−52.1	150.1	4.5	*	*	Hankard *et al*. (2007)
TG1-82	9	177.8	−75.8	55.1	6.7	*	*	Hankard *et al*. (2007)
TG1-83	9	179.3	−41.9	167.7	4.0	*	*	Hankard *et al*. (2007)
TG1-84	9	179.5	−55.8	88.2	5.5	*	*	Hankard *et al*. (2007)
TG1-85	5	168.1	−65.9	97.2	5.3	*	*	Hankard *et al*. (2007)
TG1-86	18	195.3	−41.2	138.9	6.5	*	*	Hankard *et al*. (2007)
Taatsyn Gol 1 (28.0 ± 0.8 Ma) 45.5°N, 101.1°E								
TG2-12	7	355.0	56.7	105.1	5.9	*	*	Hankard *et al*. (2007)
TG2-13	6	2.0	61.7	156.9	5.4	*	*	Hankard *et al*. (2007)
TG2-14	7	6.0	61.0	638.7	2.4	*	*	Hankard *et al*. (2007)
TG2-15	7	7.5	71.5	287.4	3.6	*	*	Hankard *et al*. (2007)
TG2-16	7	2.3	64.8	497.3	2.7	*	*	Hankard *et al*. (2007)
TG2-17	7	358.2	66.7	342.8	3.3	*	*	Hankard *et al*. (2007)
TG2-18	6	353.0	61.7	225.6	4.5	*	*	Hankard *et al*. (2007)
TG2-19	7	7.3	64.4	422.3	2.9	*	*	Hankard *et al*. (2007)

Notes: N, the number of samples in a site mean calculation; D_g, I_g, the declination and inclination of flat-lying lava flow site mean direction. k, α_{95}, the best estimate of Fisher's precision parameter and the half angle of the 95 per cent cone of confidence on the site mean direction. *Denotes if the site mean direction was used in the calculation of reference pole 3 or 4, respectively, given in Table 3a.

Table A5. Sedimentary palaeomagnetic data from Central Tibet.

Site	λ_s	ϕ_s	n	λ_p	ϕ_p	A_{95}
FHS-1: Fenghuo Shan Formation (Lin & Watts 1988)						
FH03	34.6	92.4	8	50.4	270.8	17.0
FH04	34.6	92.4	9	56.4	244.5	9.7
FH05	34.6	92.4	5	55.6	205.1	11.9
FH07	34.6	92.4	6	50.0	231.3	7.7
FH08	34.6	92.4	5	63.6	235.1	11.1
FH09	34.6	92.4	6	60.0	252.2	8.6
FE08-09	34.6	92.8	6	59.4	204.9	14.9
FE10-12	34.6	92.8	8	65.9	209.9	15.7
FE17	34.6	92.8	8	54.8	204.5	9.7
Mean	34.6	92.8	9	59.5	229.5	9.1
FHS-2: Fenghuo Shan Formation (Liu *et al*. 2003)						
Poles calculated from magnetostratigraphic sections						
GG	35.0	93.0	59	55.7	192.7	6.7
SP	34.5	93.0	51	76.0	284.7	8.3
GS	34.5	93.0	72	69.1	207.5	8.8
XLX: Xialaxiu Palaeocene-early Eocene mudstones						
(Cogné *et al*. 1999; age from Horton *et al*. 2002)						
23	32.8	96.6	6	54.9	341.0	9.1
26	32.8	96.6	6	57.4	353.8	19.2
27	32.8	96.6	5	49.0	350.8	8.3
30	32.8	96.6	6	40.0	6.2	17.3
31	32.8	96.6	7	42.6	359.1	11.8
32	32.8	96.6	5	40.2	351.9	10.6
Mean	32.8	96.6	6	47.6	354.4	7.9

Notes: λ_s, ϕ_s, the latitude and longitude of the sampling site; N, the number of sites used to calculate the site mean; λ_g, ϕ_g, A_{95}, the latitude, longitude and half angle of the 95 per cent cone of confidence of the palaeomagnetic pole for each study.